

MPLS

Multi-protocol label switching

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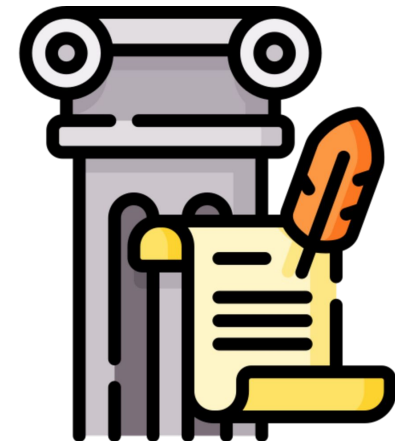


Introduction to MPLS



MPLS history

- A great portion of packets traversing the internet today goes through an **MPLS network**
 - MPLS is one the most important protocols running in the **core** of the network
- A protocol similar to MPLS was proposed in **1996**, but it could be applied to ATM only
 - It never had success
- Cisco revised the proposal and made it applicable to **all layer-3 protocols**
 - **Multi-protocol** label switching
 - Back then, there were **multiple layer-3 protocols** (now only IPv4 and IPv6)



Back in the '90s

- Communication Technology (CT) world
 - Telcos operators, telephone calls
 - Based on **circuit-switching** mechanism
 - I create an **end-to-end circuit** and then I start the call
 - Main protocols: **SONET/SDH**
- Information Technology (IT) world
 - Internet, Computers, www, etc.
 - Based on **packet-switching** mechanism
 - I send the packet **independently** and with **hop-by-hop routing** decisions
 - Main protocols: **IP**
- Information and Communication Technology (ICT) thanks to MPLS



MPLS goal

- MPLS was proposed as (and still is) the enabling technology for the **New Broadband (IP) Public Network**
 - **Private Network** → Polito network carries Polito traffic only
 - **Public Network** → Telecom Italia, Fastweb, etc. networks carries traffic of everyone and makes money over this
- IP has **no QoS guaranteed** (best effort)
 - IP + MPLS → made existing public IP network suitable for **selling services with requirements**



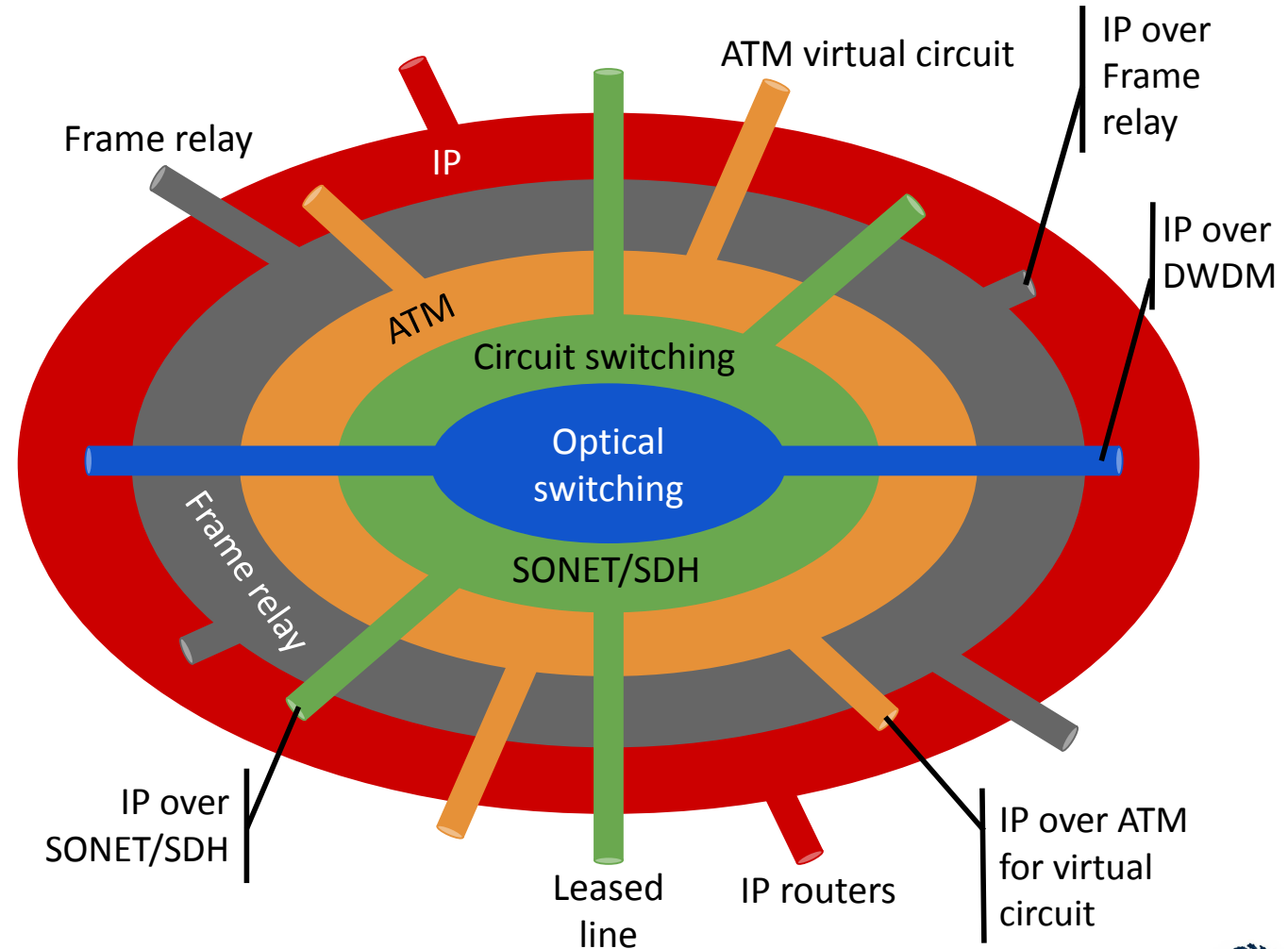
From ATM to MPLS

- CT protocols were **worldwide established**
- IP was a **brand-new protocol**
 - Asynchronous Transfer Mode (**ATM**) architecture proposed by CT
 - Reuse existing CT infrastructure with IP (**IP over ATM**)
- ATM architecture had several issues
 - **Great overhead** needed to construct the virtual circuit → made IP extremely slow
 - ATM routers very complex and **very expensive**
 - The more IP got spread, the more ATM issues become relevant, the more everyone realized **ATM was not the solution**
- MPLS proposes itself as the solution for **both CT and IT worlds**
 - MPLS to substitute ATM and **enhance IP with QoS** to be guaranteed → ok for IT
 - MPLS could be **used also on old and expensive ATM routers** → ok for CT
- MPLS was **immediately adopted**



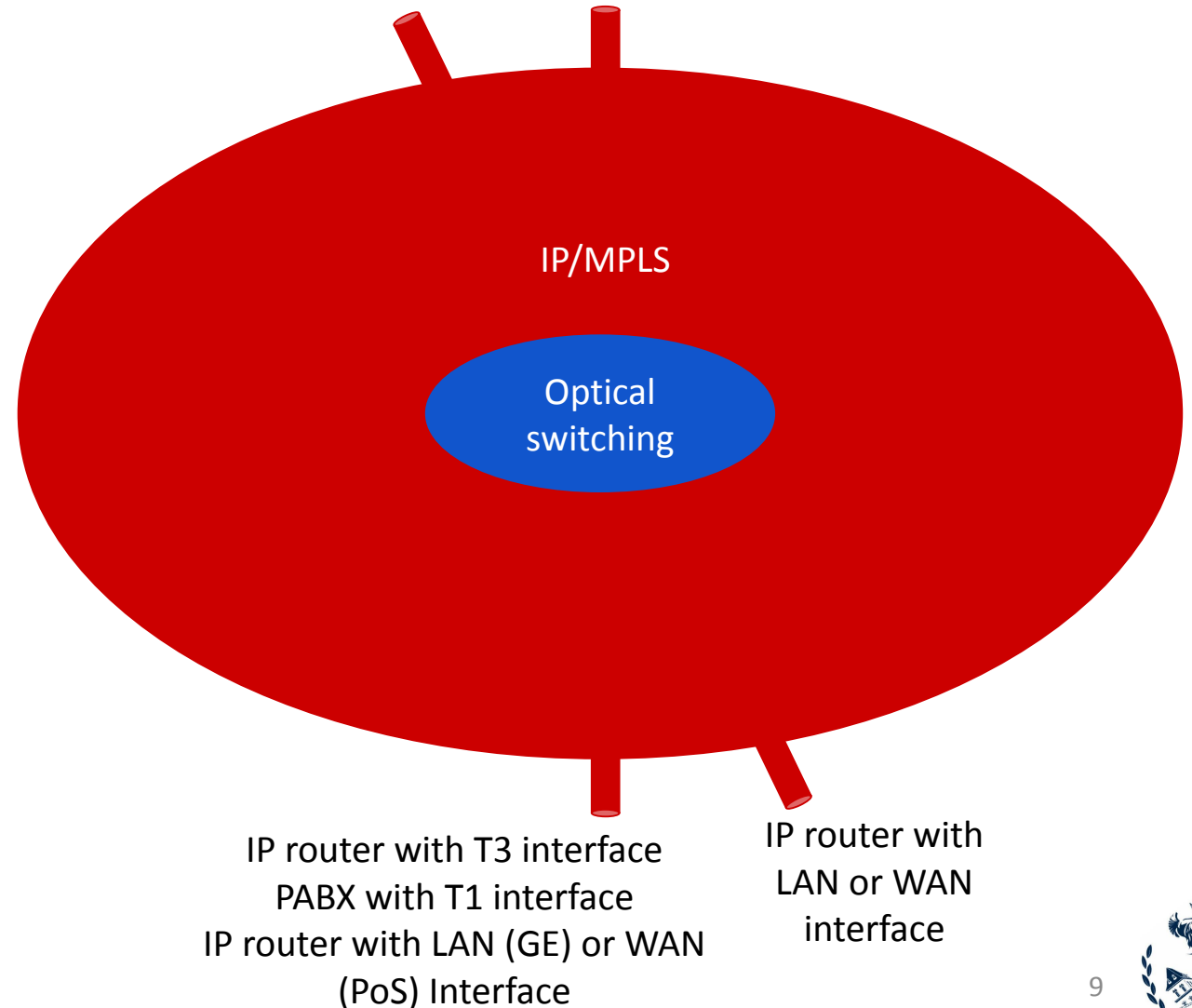
The “onion” that made telcos “cry”

- Situation of a network in late '90s
 - Optical switches for PHY layer
 - Circuit switching + ATM for creating a virtual circuit for calls
 - Frame relay to reuse the same structure for packet switching
 - IP for packet switching
- Extremely complex architecture
 - CT (Communication Technology) vs IT (Information Technology)



MPLS architecture

- Keep optical switching as PHY layer
- Advantages for CT world
 - Get rid of all old SW protocols
 - Maintaining the same QoS of the old SW protocols
- Advantages for IT world
 - Guarantees on QoS with IP, which was not possible to have before
 - Possibility to sell services based on QoS
- Suited for both CT and IT
 - ICT (Information AND Communication Technology)
 - Immediately adopted



The idea

- We have IP packets with IP destination address
 - IP dst addr = 32 bit in IPv4
 - IP dst addr = 128 bit in IPv6
- Routing in IP is done with IP dst addr
 - Longest Prefix Matching (LPM)
 - LPM discriminate on IP dst addr (single users) or aggregation on them (geographical areas or institutions)
- MPLS idea is to add a label in front of the packet
 - 20 bits-long label
 - MPLS labels is not associated to anything related to the IP packet, so it can discriminate on anything
- We use MPLS labels for packet forwarding instead of IP dst addr



Why?

- MPLS enhanced significantly the performances of IP routing
 - LPM is a time-wise and resource-wise complex operation
 - Often requires to scan all the IP routing table
- Simpler routing tables → faster lookup
 - Routing tables with (Label; next hop) → label as an index
 - One-to-one match with routing tables
 - Label 5 = 5th entry in the routing table
- This was true in the '90s
 - Now TCAM routers exist, allowing a fast LPM lookup
 - TCAM routers are very expensive
 - MPLS is a SW solution → cost (almost) nothing
 - MPLS is still the main solution today!!



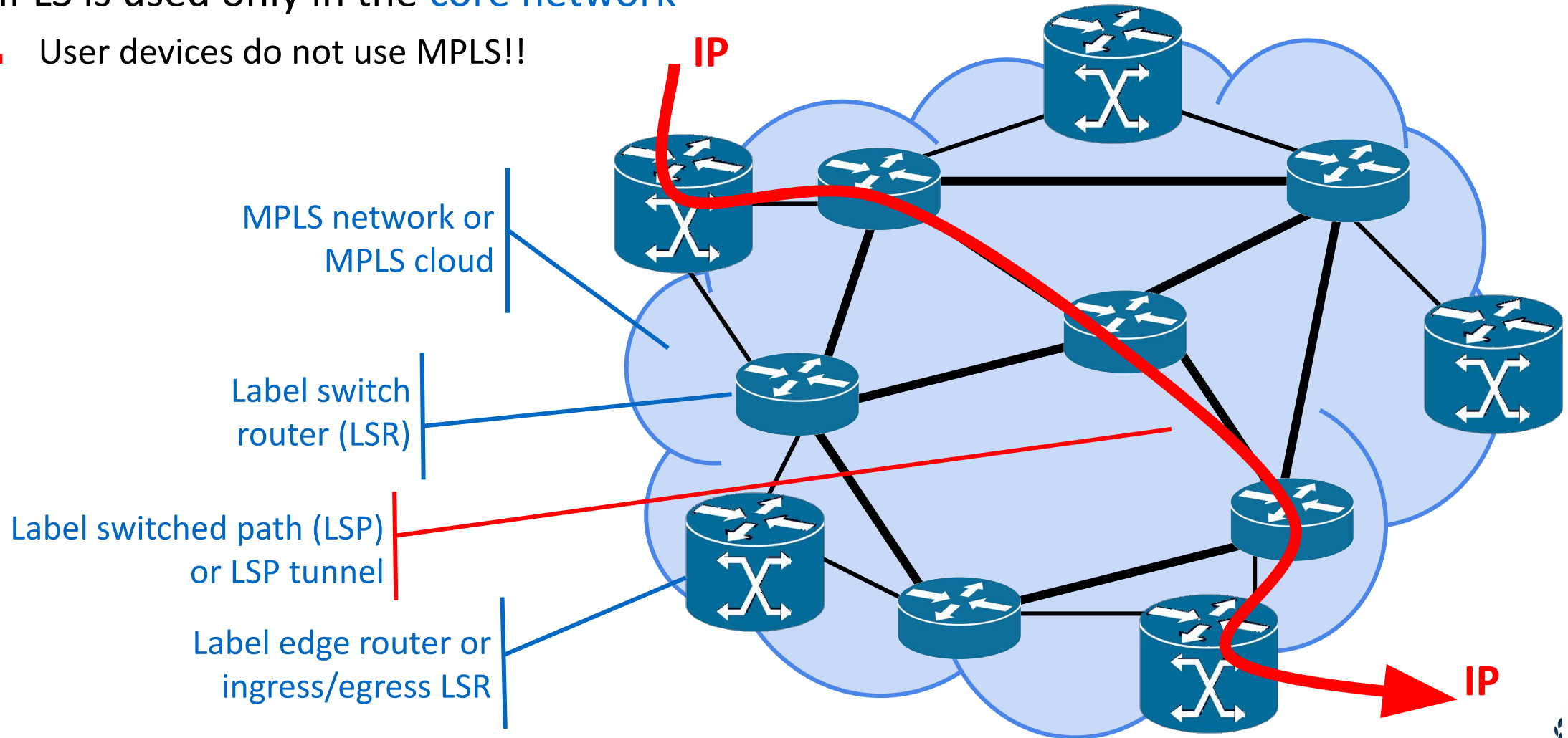
Why?

- Labels can be associated to anything
 - Allows Traffic Engineering
- Traffic of different hosts in the same network can be very different
 - Traffic of server Portale Didattica VS traffic public IP server of my lab
 - With routing based on IP dst addr I can or discriminate on all dst hosts (extremely long tables), or on aggregation of them (I treat them all the same)
 - With Traffic Engineering I can choose ad-hoc routing (less congested routes for high traffic servers)
- MPLS introduces a connection-oriented paradigm in IP networks
 - IP is connectionless → Connections are on layer 4
 - With MPLS I can have similar idea of circuit switching with IP networks
 - Allows service providers to sell QoS-based services on the public network

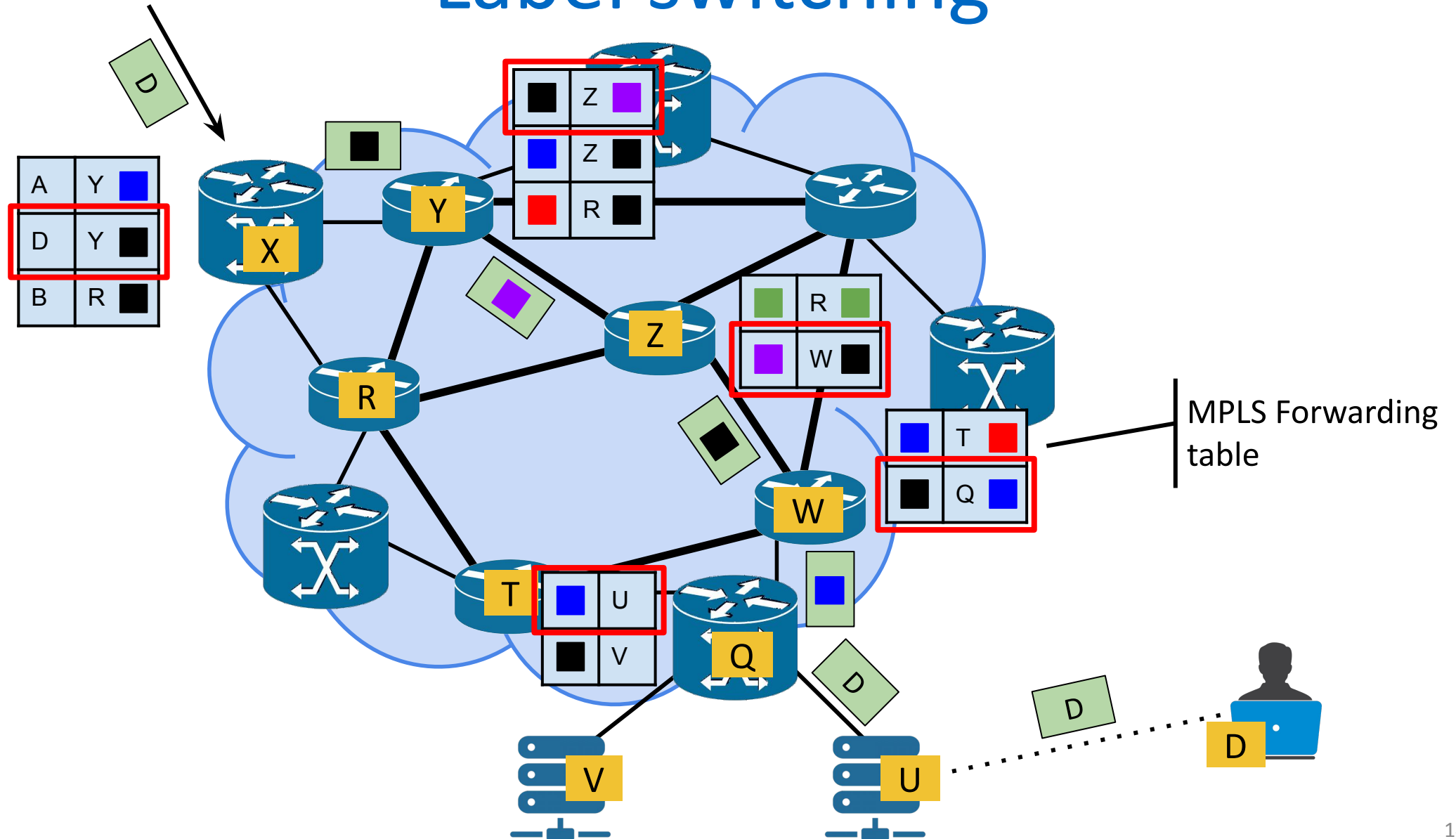


MPLS network

- MPLS is used only in the **core network**
 - User devices do not use MPLS!!

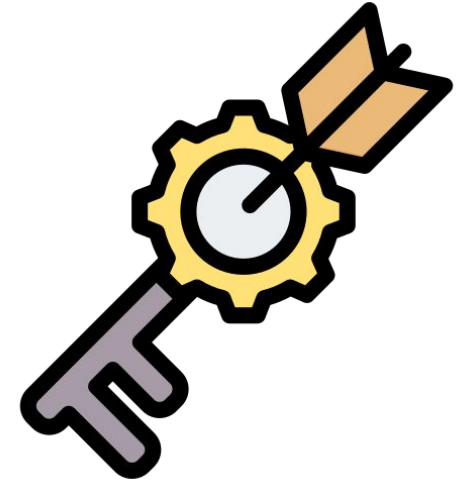


Label switching



MPLS key elements

- MPLS “header”
 - Contains the **MPLS label**
- Protocols for label distribution
 - Signaling protocol for **allowing LSR to create the LSP**
 - IP routing is only based on destination addresses
- Enhanced routing protocols
 - Same idea of IP routing, but enhanced with some **constraints in choosing path**
 - IP → shortest path
 - MPLS → shortest path **with QoS constraints**
 - “Find a shortest path **whose links are used at most 50%**”
 - “Find a shortest path **whose links have a minimum capacity**”
 - “Find a shortest path **whose links are independent from links of another path**”



Other MPLS versions

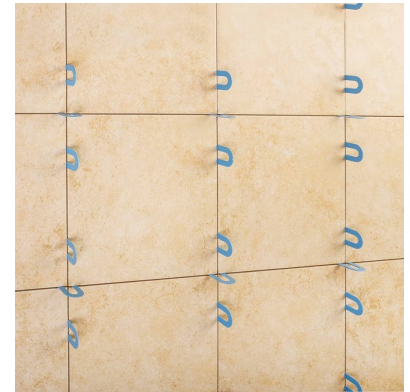
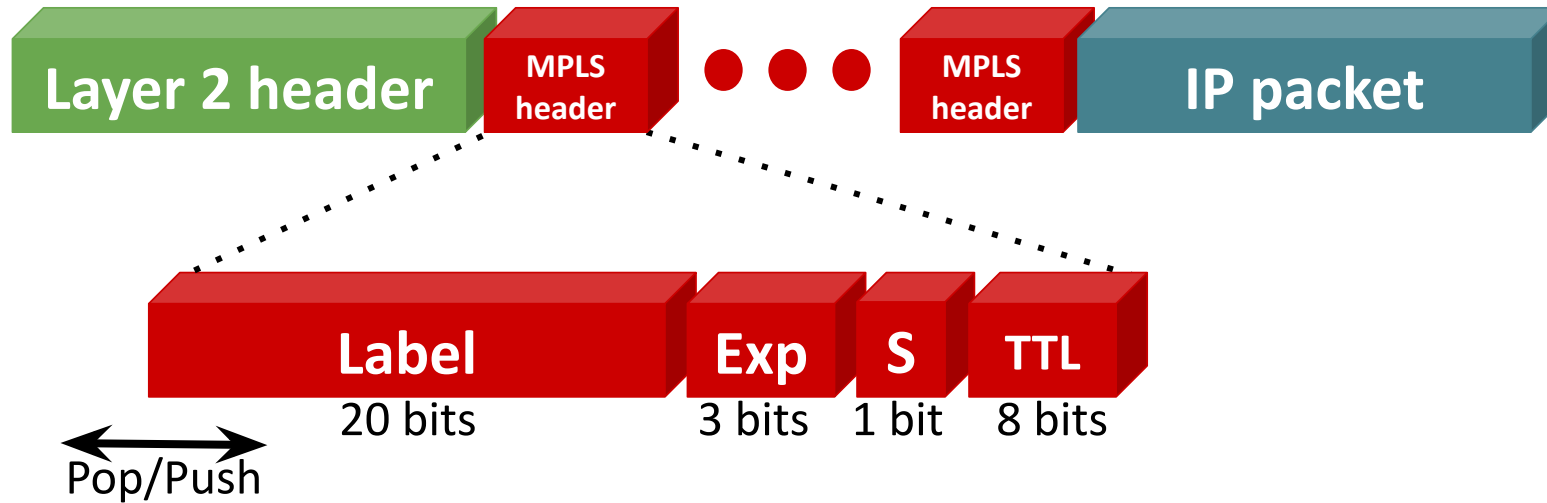
- MP λ S ([Multi-Protocol Lambda Switching](#)) for optical switching
 - Decide which frequency to use is similar to MPLS problem of what label to use
- G-MPLS ([Generalized MPLS](#)) suited for:
 - Packet switching $\rightarrow$ IP
 - Cell switching $\rightarrow$ ATM
 - Circuit switching $\rightarrow$ SONET/SDH
 - Lambda switching $\rightarrow$ Optical routing
 - “Anything” switching due to a [control plane compatible with multiple protocols](#)



MPLS header



Shim header



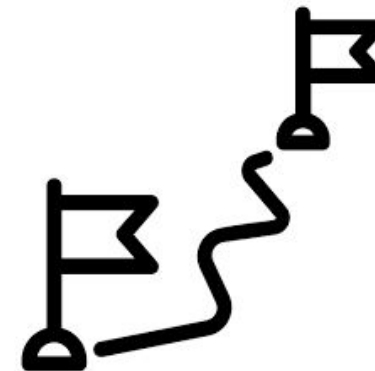
- Fields of MPLS header are **popped or pushed as a stack** (LIFO) from the left
 - Exp: experimental bits, usually used for **Classes of Service (CoS)**, aka priorities
 - S: **bottom of stack**, to signal the presence or not of a next MPLS header block
 - TTL: **Time To Live**
- If layer 2 is **ATM or Frame relay**, MPLS label can be put inside a layer 2 field
 - This choice made old ATM and Frame relay routers **compatible** with MPLS!!

LSP setup: label and path selection



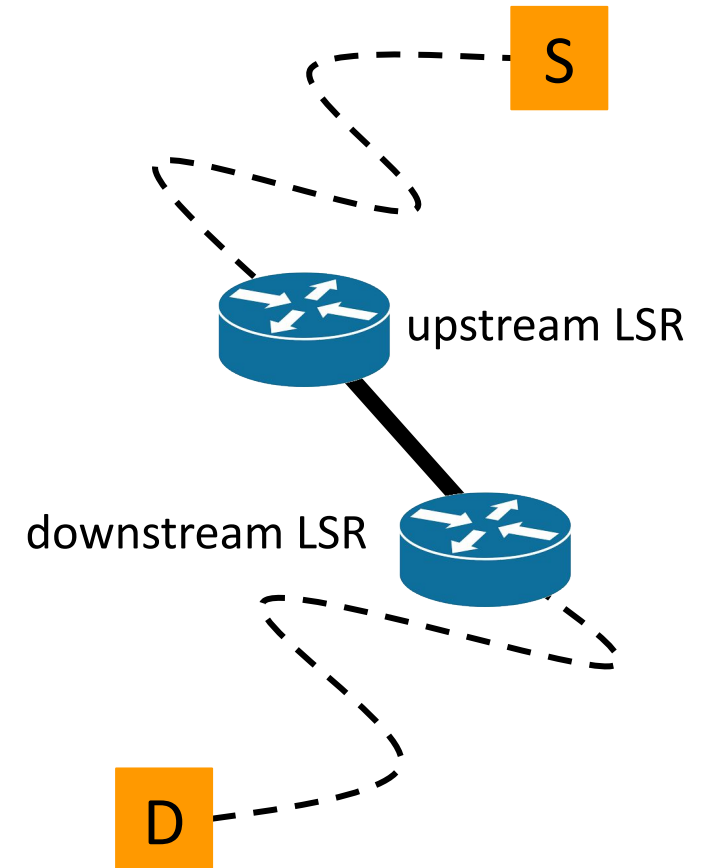
LSP setup

- Forwarding Equivalence Class (**FEC**) is a set of packets which are treated the same way by each LSR
 - They will follow the **same path** in the MPLS network (with the **same direction**)
 - They will have the **same labels** across the path
- LSRs assign labels to FECs (**create LSPs for each FECs**)
 - IP aggregates flows in dst addr groups (**subnet**)
 - MPLS can also defines **more fine-grained parameters** for the FEC (RX/TX LSR port, src IP addr, etc.)
- LSP setup is performed in 3 steps
 - **Label binding**
 - **Label mapping**
 - **Label distribution**



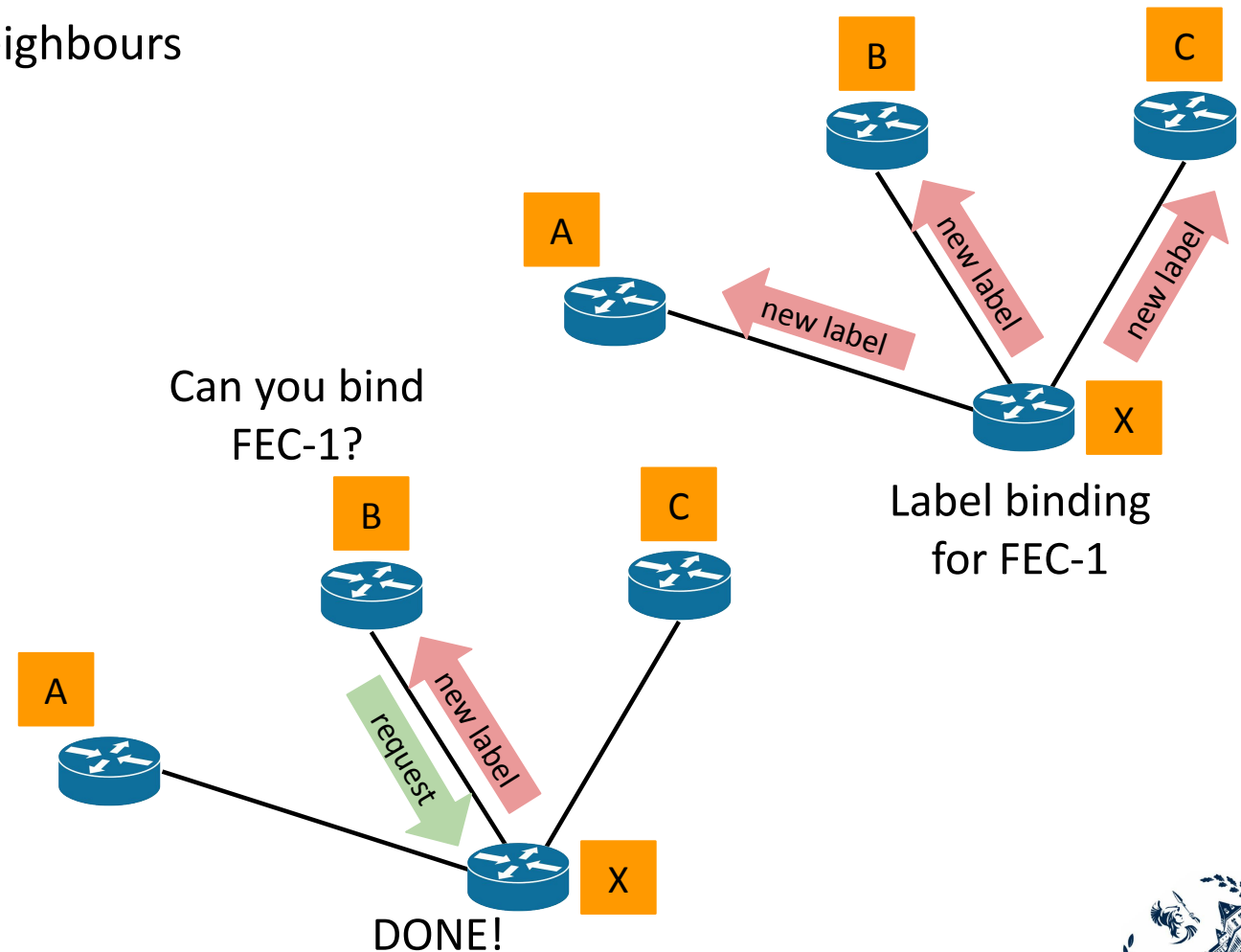
Label binding

- Label binding → choosing a label for a FEC
- In MPLS label binding is always done by downstream LSRs
 - In a link between two LSRs, the LSR closest to the destination is the downstream LSR
 - The downstream LSR performs label binding → choose the label
 - The downstream LSR are the one performing the lookup in the table
 - They will choose labels which are easier (faster) to lookup for them
- Downstream LSR will inform the upstream LSR of the link regarding the chosen label
 - This is done in the label distribution phase
- Label binding can be unsolicited or on-demand
 - Unsolicited: an LSR independently decides to bind a label to a FEC
 - On-demand: someone ask the LSR to bind a label to a FEC



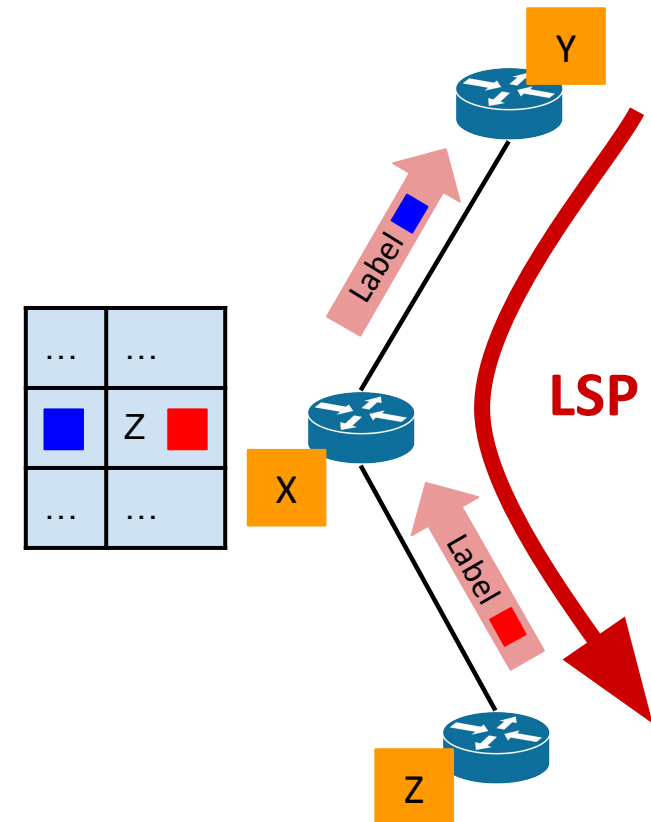
Label distribution

- Notification from a LSR to another that a label was bound
 - Performed from a downstream LSR to its neighbours
 - Labels must be unique on the link!!
 - Labels are changed every link of the LSP
- If the label binding was **unsolicited**:
 - All LSR neighbours are notified
- If the label binding was **on-demand**:
 - Only the demanding neighbour is notified

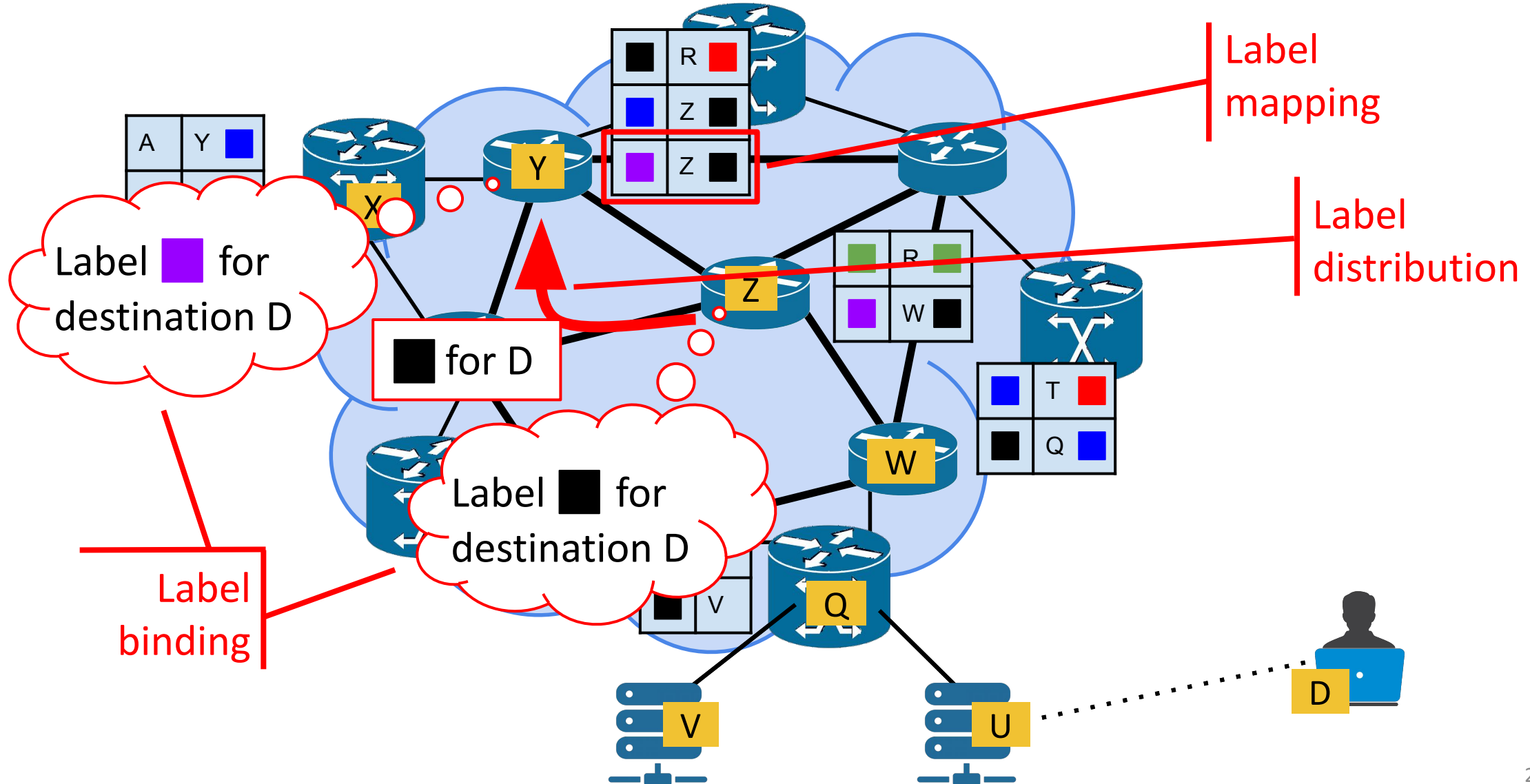


Label mapping

- Association between:
 - Input label (just chosen by the current LSR during label binding)
 - Output label (chosen by the downstream LSR wrt the current one in its label binding)
 - Next hop (based on routing)
- Current LSR can perform label mapping **only** after the label distribution of downstream LSRs
- Label mapping actually **creates the LSP**



Sample LSP setup



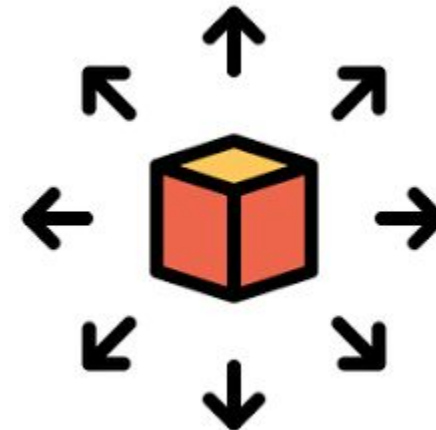
Static/dynamic label binding

- Label binding (and mapping) can be decided **statically**
 - By **network admin** at the setup phase
 - Not practical because **cannot react to network changes**
 - It is **not scalable**
- It can be **dynamic and data-driven**
 - It can be **data-driven** → I receive a **new packet for which I don't know the label**, then I start the label binding
- Or **dynamic and control-driven**
 - **Control-driven and topology-based (unsolicited)** → I discovered a **new next hop via routing protocols**, then I start the label binding for that next hop
 - Similar to IP and **not good for traffic engineering**
 - **Control-driven and explicit (on-demand)** → I receive a **control message to signal the binding start**
 - Started by the ingress LSR for all the LSRs of the path
 - Similar to **connection-oriented** protocols



Label distribution protocols

- Three alternatives for **label distribution** (incompatible one each other)
- **Routing protocol** (e.g., Border Gateway Protocol – **BGP**)
 - We are going to use BGP anyway for routing, so we can use it also for label distribution
 - Can be used **only for content-driven and topology-based binding**
- **Label Distribution Protocol (LDP)**
 - Made **on purpose for MPLS** by Cisco (**proprietary** protocol)
 - Nowadays used **only on Cisco routers**
- **Resource reservation protocol (RSVP)**
 - Used **also for reservation of resources and QoS**
 - Generic protocol **not proprietary**



MPLS routing protocol



Routing protocols in MPLS

- Packets are forwarded based on the LSP
- But routing protocols are used to create the LSP
 - In the label mapping phase
 - [Input label; Next hop; Output label] entries in the MPLS forwarding table
 - Next hop is determined by routing protocols
- Routing protocols impact the label mapping phase



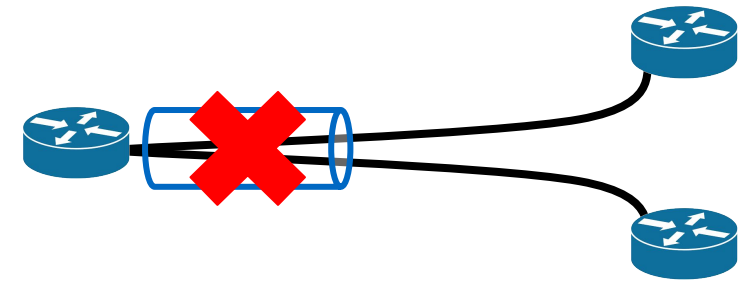
Routing protocols in IP

- In MPLS we can reuse (re-adapt) some routing protocols existing for IP
 - Not all, but just the more scalable which are suited for very large networks
- Open Shortest Path First (OSPF)
- Intermediate System to Intermediate System (IS-IS)
- Border Gateway Protocol version 4 (BGP-4)
- They all carry topological information (about the links, the routers, links speed, etc.)



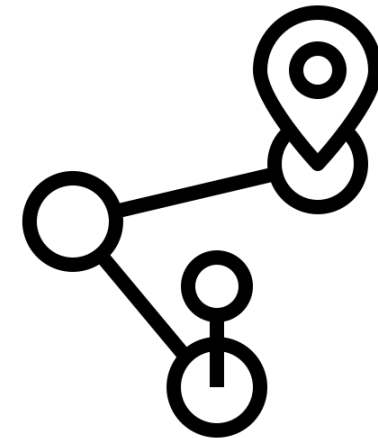
Enhanced routing protocols for MPLS

- In MPLS we **re-adapt** the same IP protocols, **enhancing** them
 - We use them to carry information **to constraint routing decision** (**constraint data**)
- Not just shortest path
 - “Find a shortest path **whose links are used at most 50%**” → Link utilization
 - “Find a shortest path **whose links have a minimum capacity**” → Link capacity
 - “Find a shortest path **whose links are independent from links of another path**” → Link dependencies for fault recovery
- **Constrained data** are important for two reasons
 - **Traffic engineering**
 - Backup LSPs for **fast fault recovery**
- New **enhanced routing protocols** for traffic engineering
 - **OSPF-TE** (OSPF supporting Traffic Engineering)
 - **IS-IS-TE** (IS-IS supporting Traffic Engineering)



Routing modes

- In MPLS there are two routing modes
 - Hop-by-hop routing
 - Explicit routing
- Hop-by-hop routing
 - Every LSR independently looks at its routing table and picks the next hop to the destination
 - There is no global decision for the LSP
- Explicit routing
 - One LSR decides the whole path

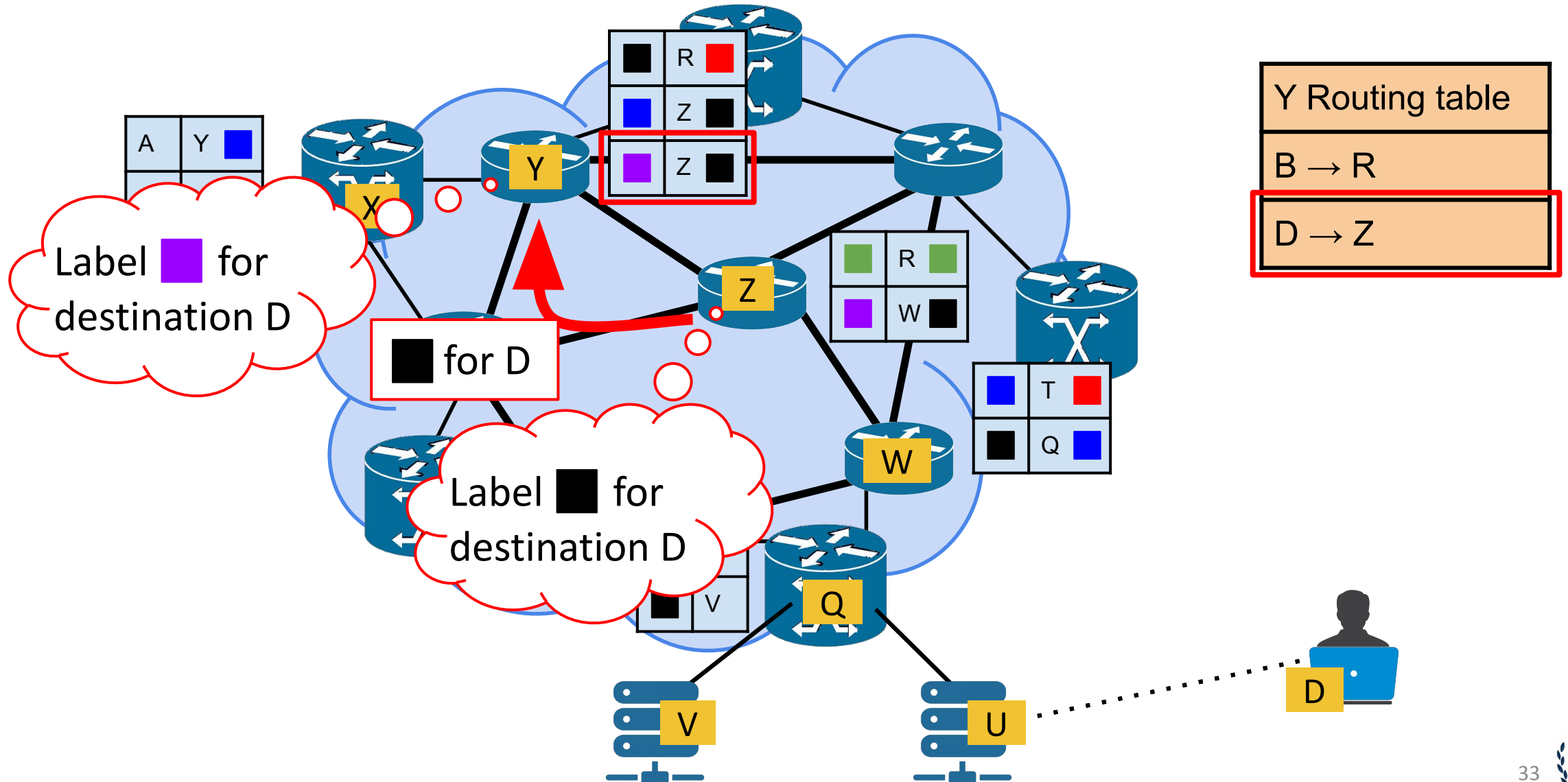


Hop-by-hop routing

- When a LSR has performed **label binding** and has received the output label from the neighbours LSRs via **label distribution**
 - Then the LSR looks at its **IP routing table**
 - It determines the **next hop**
 - It can create the **label mapping**
- It will provide the very **same routing result of pure IP** (few advantages from MPLS)
 - LSR will be faster in the lookup thanks to labels
 - But **no Traffic Engineering** can be performed!!



Hop-by-hop routing example



Explicit routing

- Used for **Traffic Engineering**
 - With Traffic Engineering, **constraint paths** are used
 - Normal IP routing (like hop-by-hop routing) are not valid anymore with constraint paths
 - Distributed constraint-based routing is impossible cause there can be **conflicting constraints**
 - **Explicit constraint-based routing**
- Only **one LSR chooses for all** the other LSRs of the LSP
 - Usually the **ingress LSR** performs explicit routing
 - It **chooses all the hops of the LSP** based on constraints
- The ingress LSR will then **inform all the other LSRs** of the LSP about the path
 - It will do that thanks to **enhanced versions of label distribution protocols** (LDP or RSVP)
 - Constraint-based Routing LDP (**CR-LDP**)
 - RSVP for Traffic Engineering (**RSVP-TE**)

Advantages of explicit routing

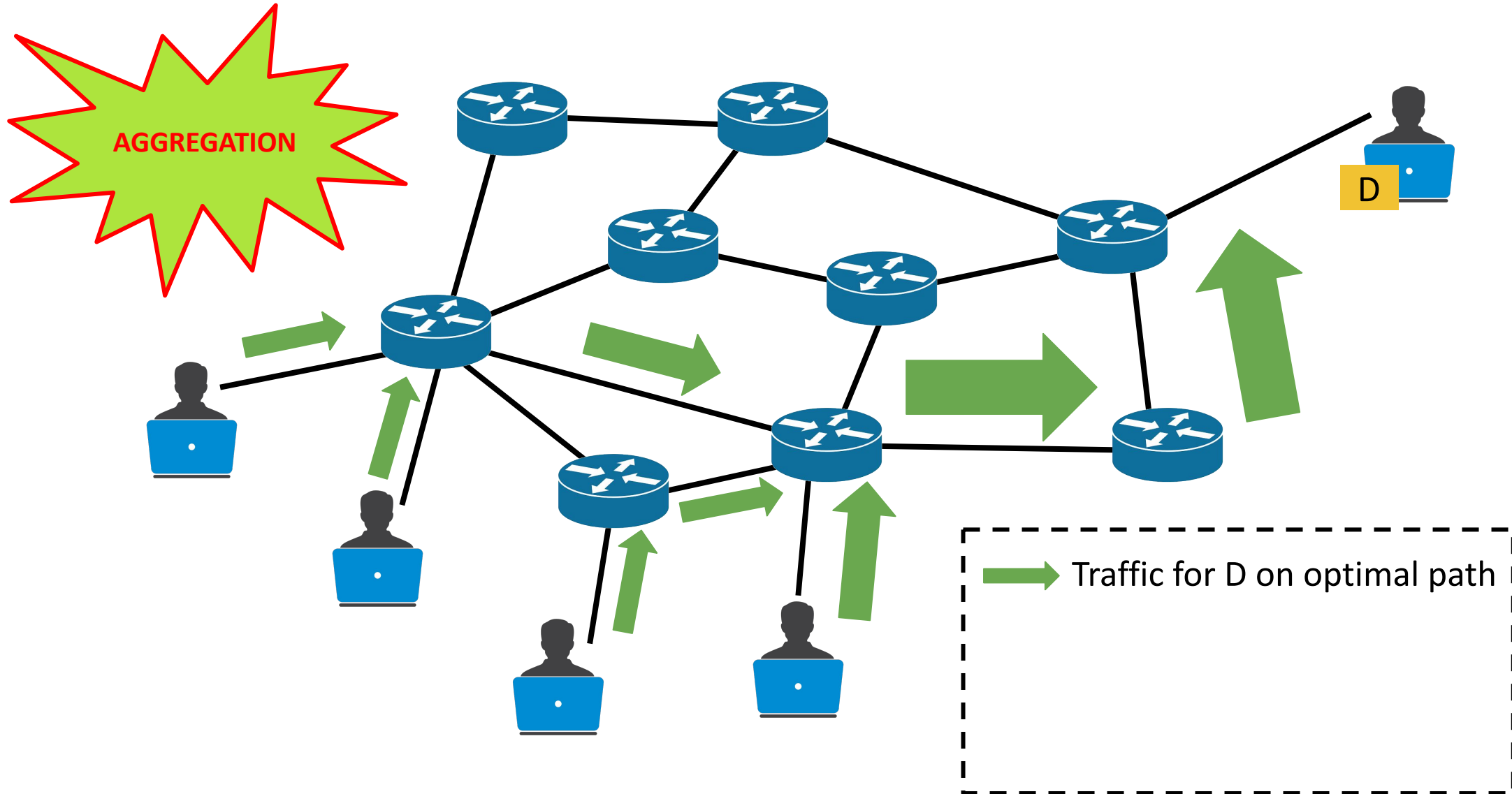
- Traffic Engineering is possible
- Guarantees on QoS (theoretically)
 - Not yet supported today in MPLS
 - Only way to guarantee QoS is to reserve resources on all the links
 - Service Providers don't want to reserve resources for everyone
 - Resource reservation only for specific contracts for big agencies
- Classes of Service (CoS) distinction
- Fast fault recovery
 - Fault recovery in less than 50 ms
 - Not possible with traditional routing protocols based on IP



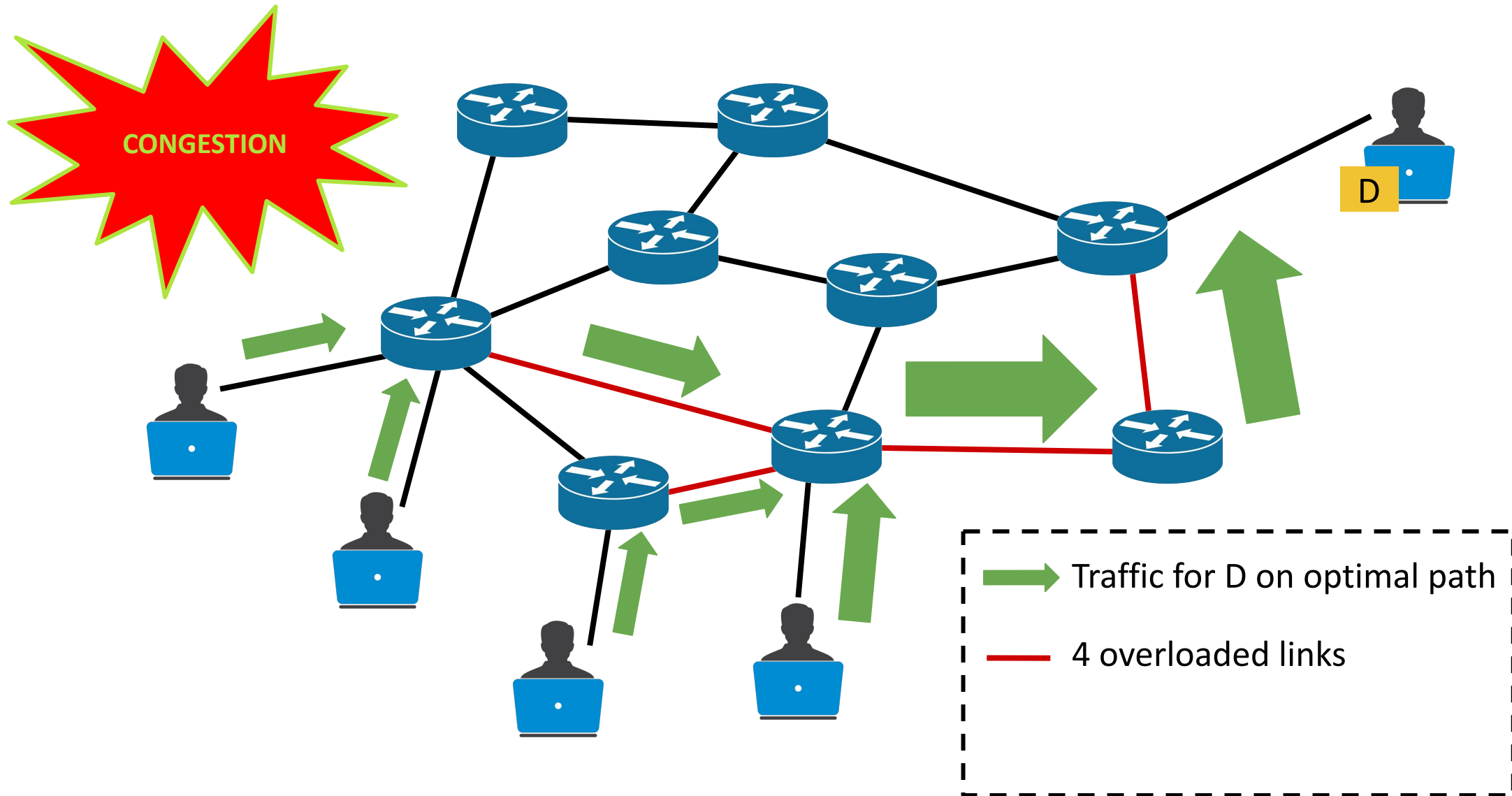
Traffic Engineering



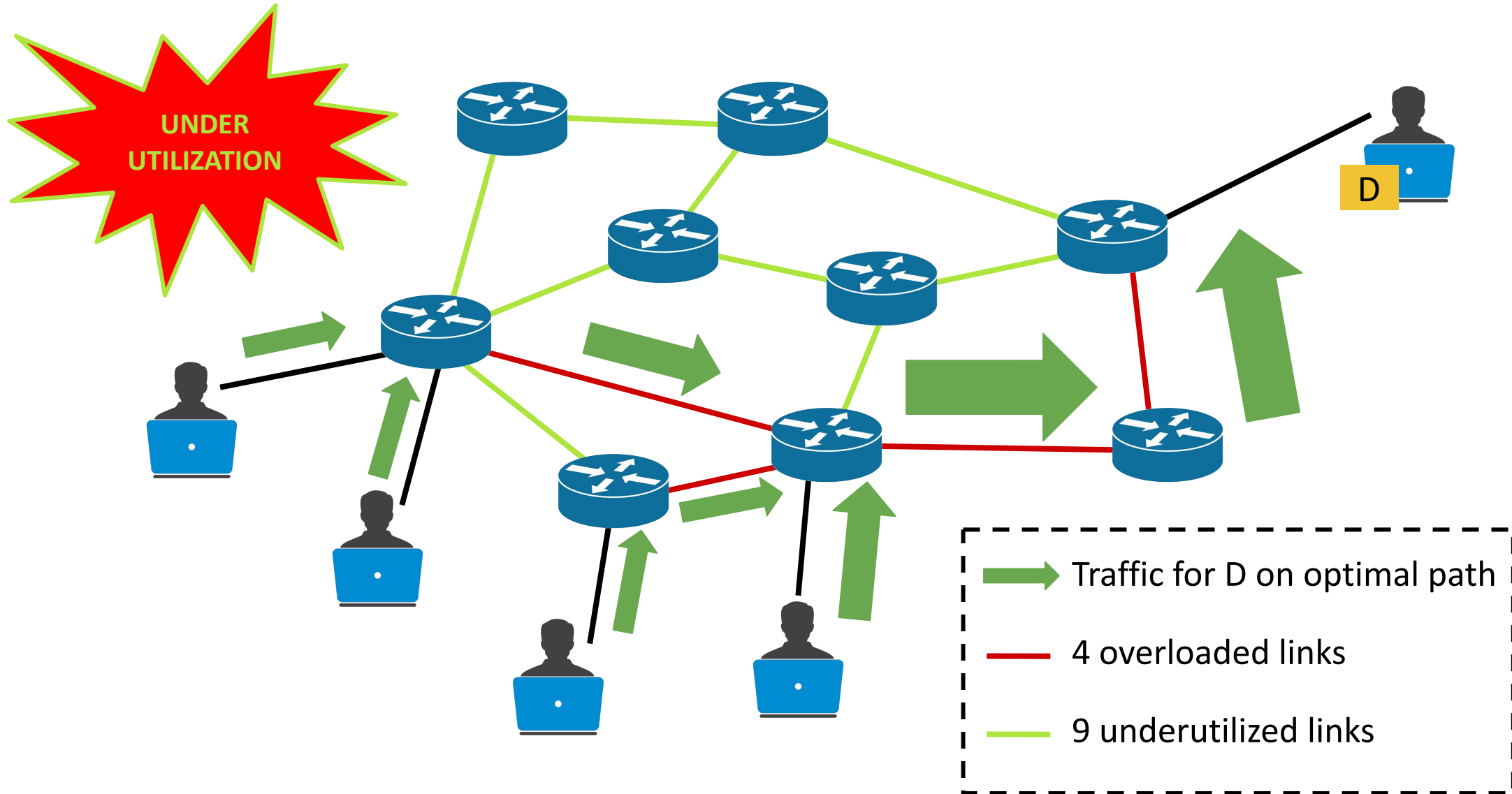
Traditional IP routing



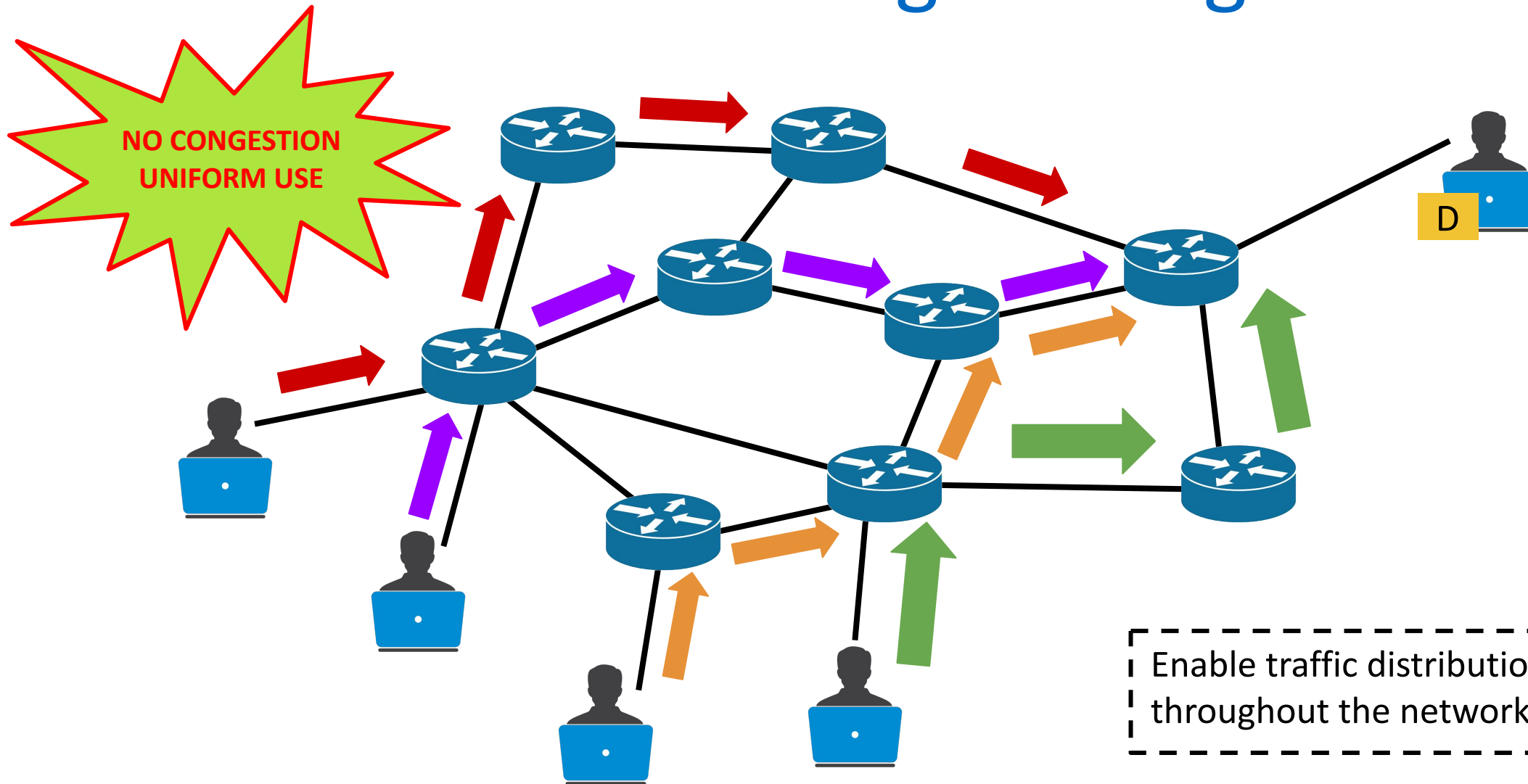
Traditional IP routing



Traditional IP routing



Traffic Engineering

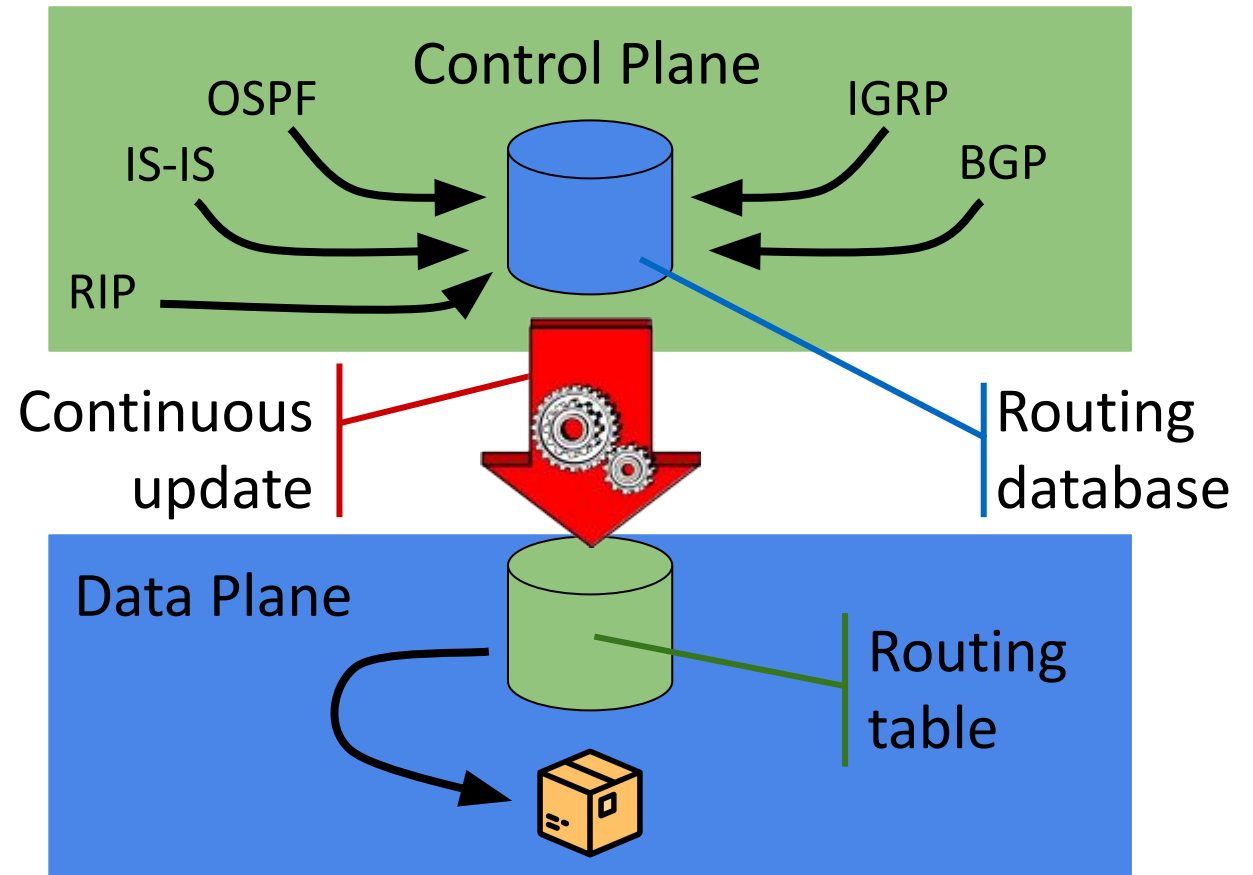


IP router Control Plane & Data Plane

- All routers are composed by a **Control Plane** and a **Data Plane**

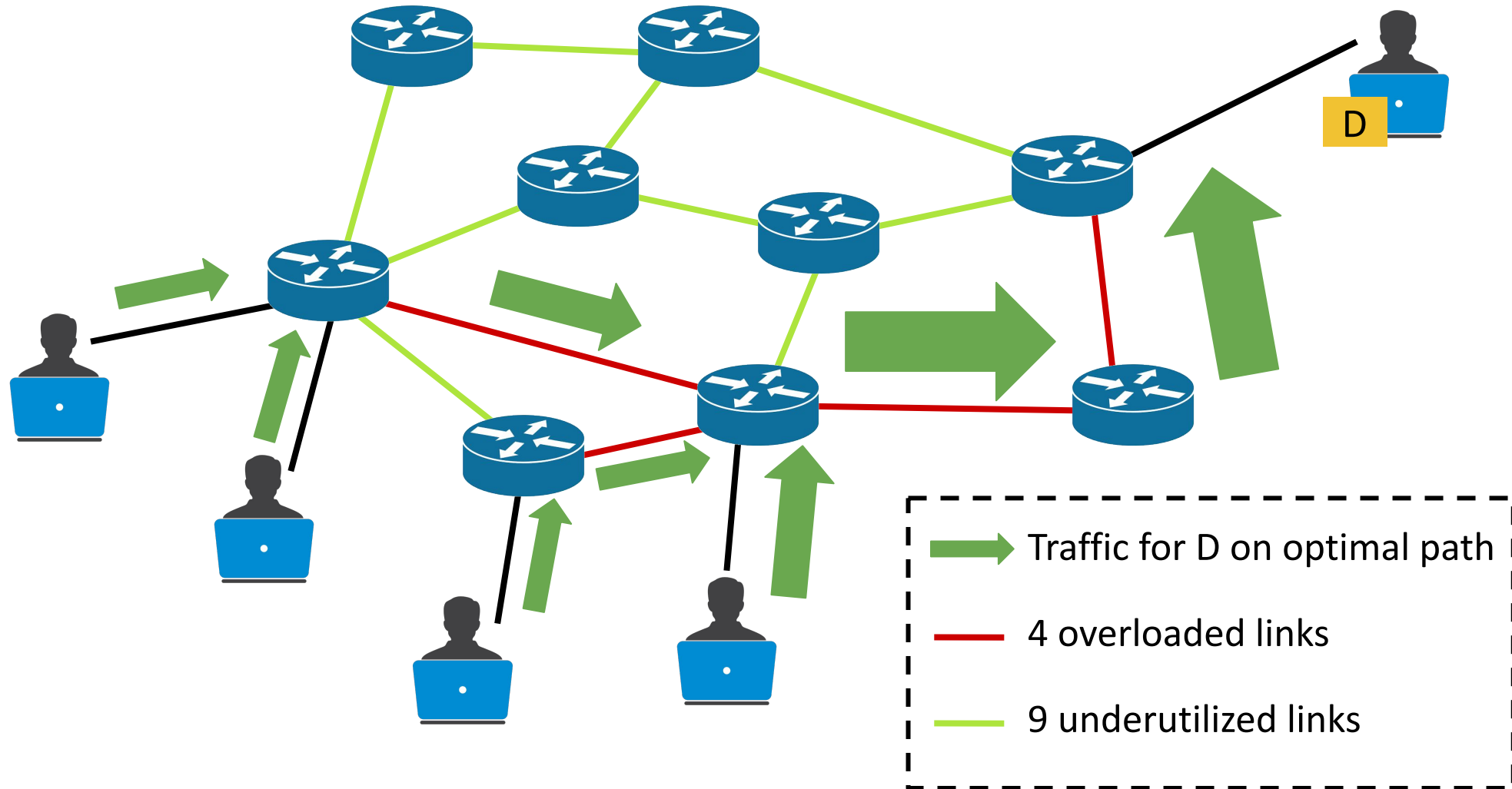
- Control Plane contains the **routing protocols** and the **routing database**
- Data Plane is where **packets are processed** based on the dst addr and the **routing table**
- Control Plane is implemented in **software**
- Data Plane is implemented in **hardware**

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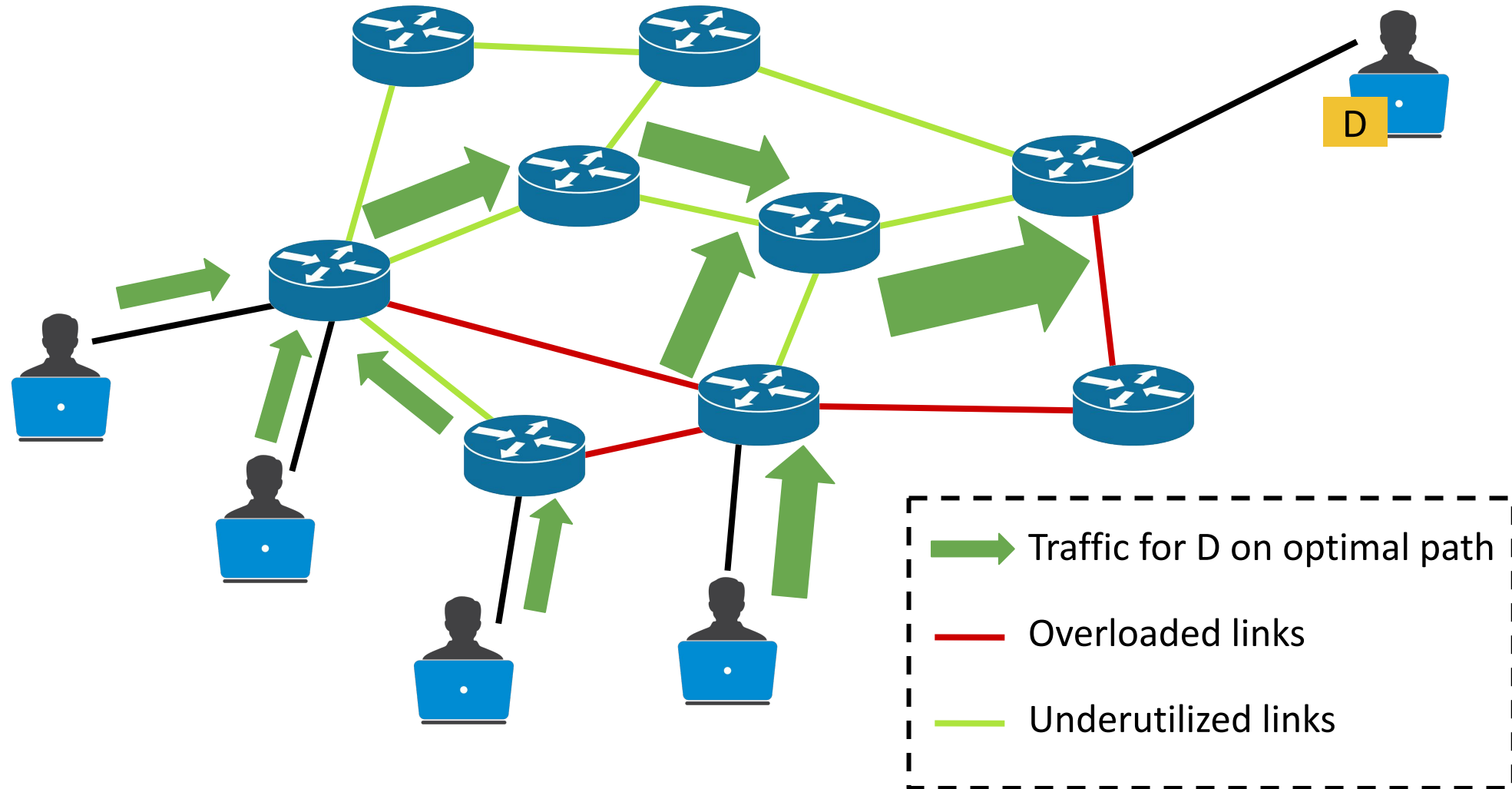


- In IP routers every time there is a new or better destination, **the routing table is recomputed**

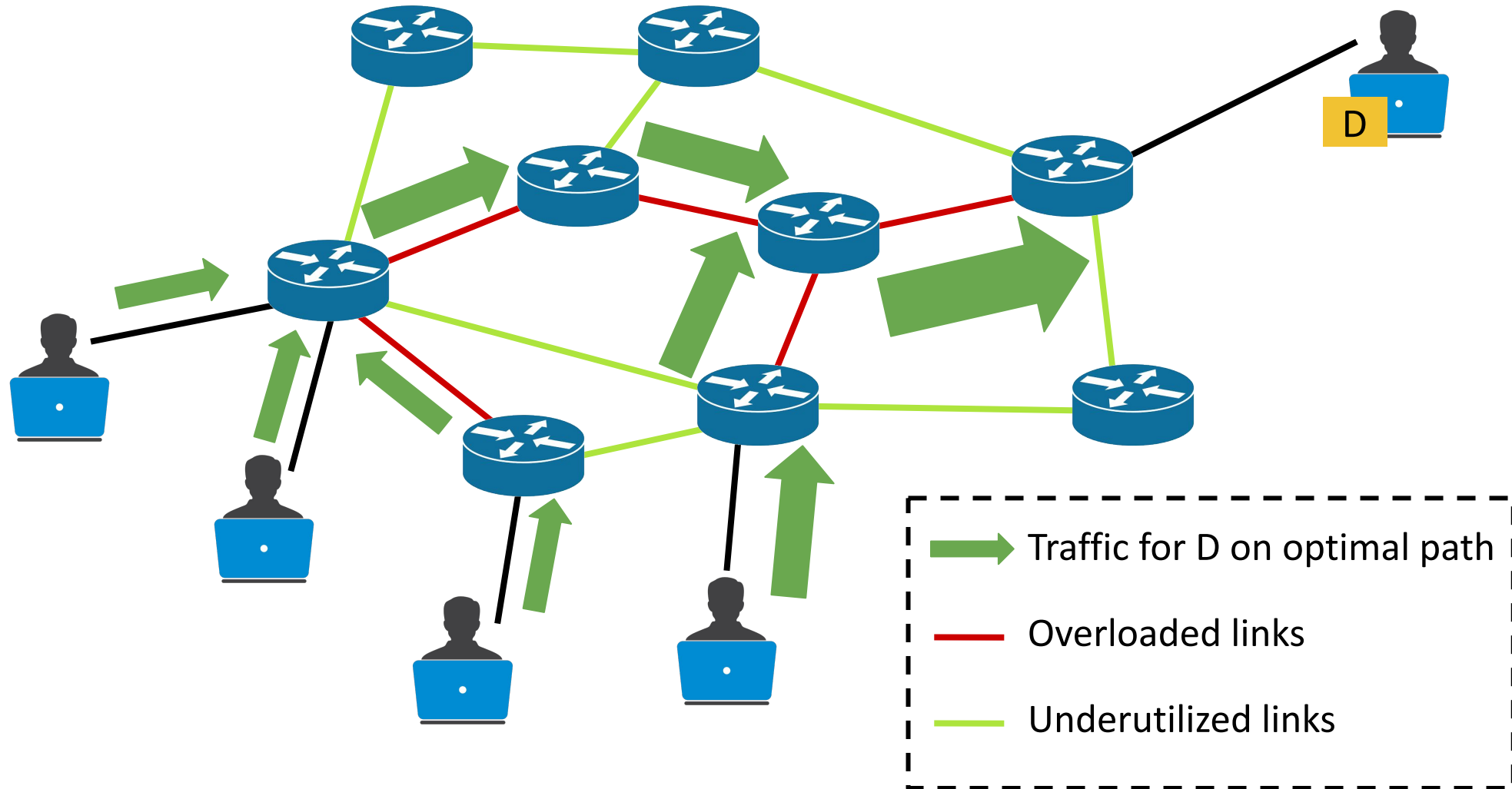
Can we do the same with IP routing?



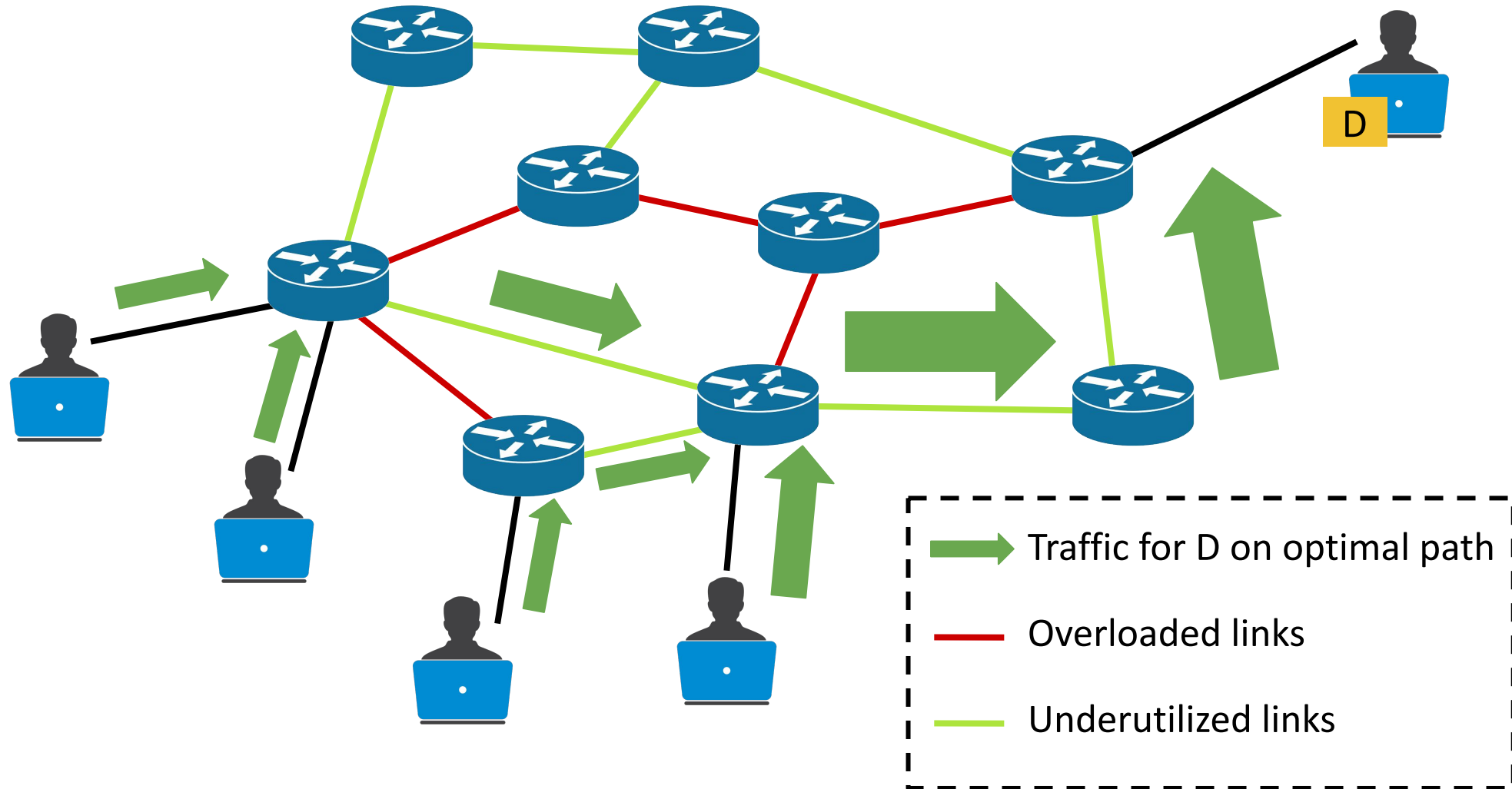
Can we do the same with IP routing?



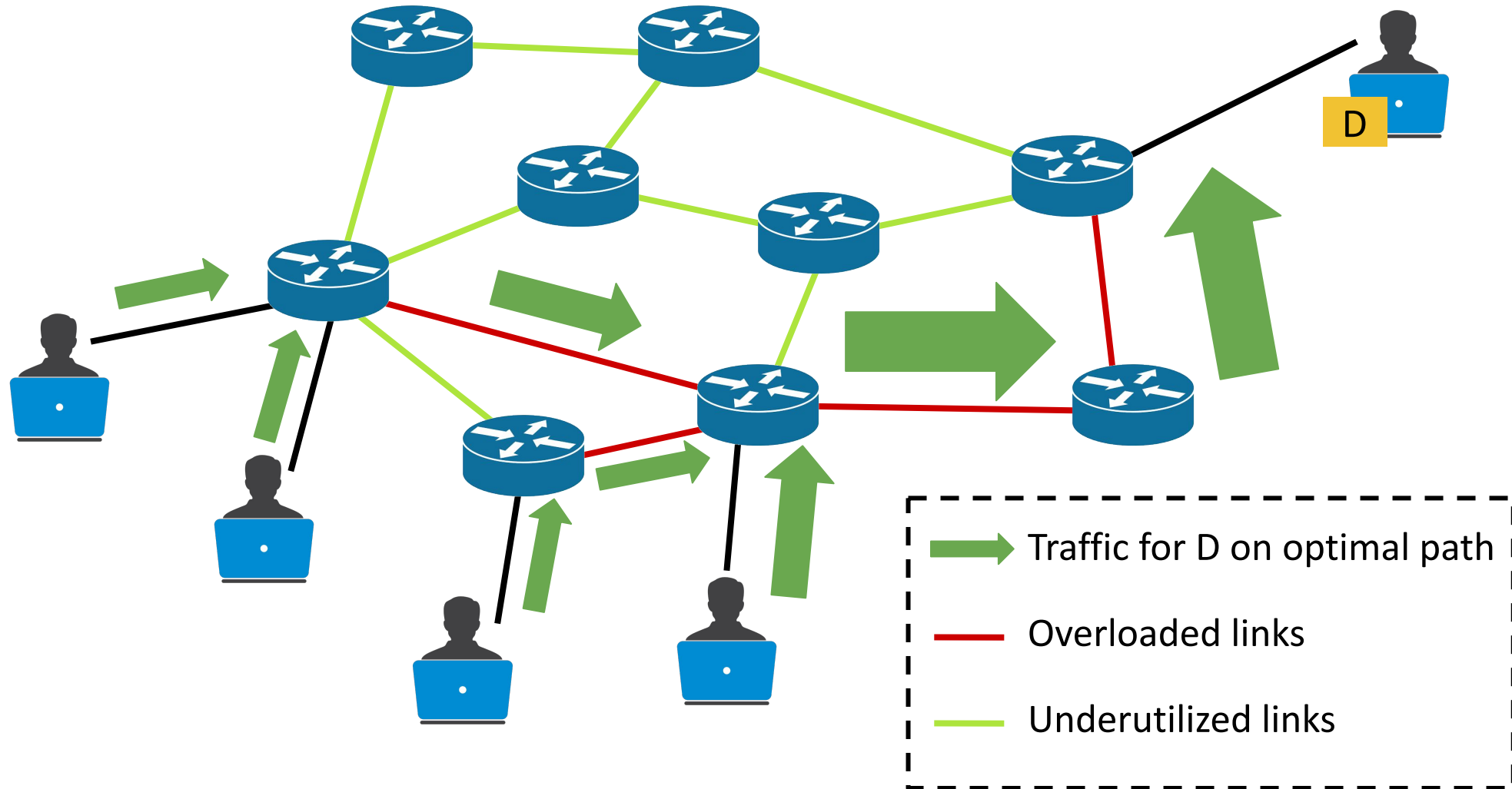
Can we do the same with IP routing?



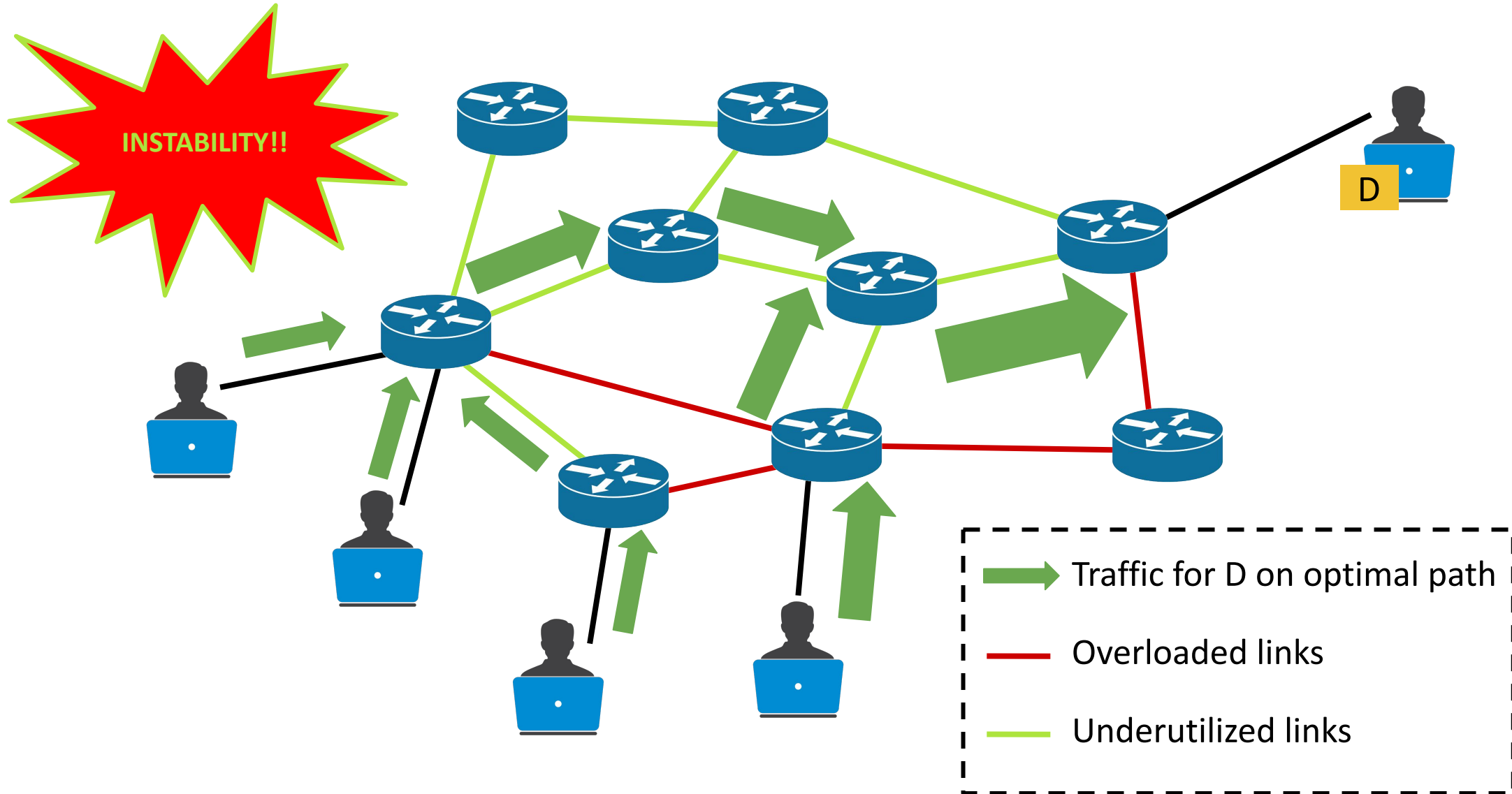
Can we do the same with IP routing?



Can we do the same with IP routing?



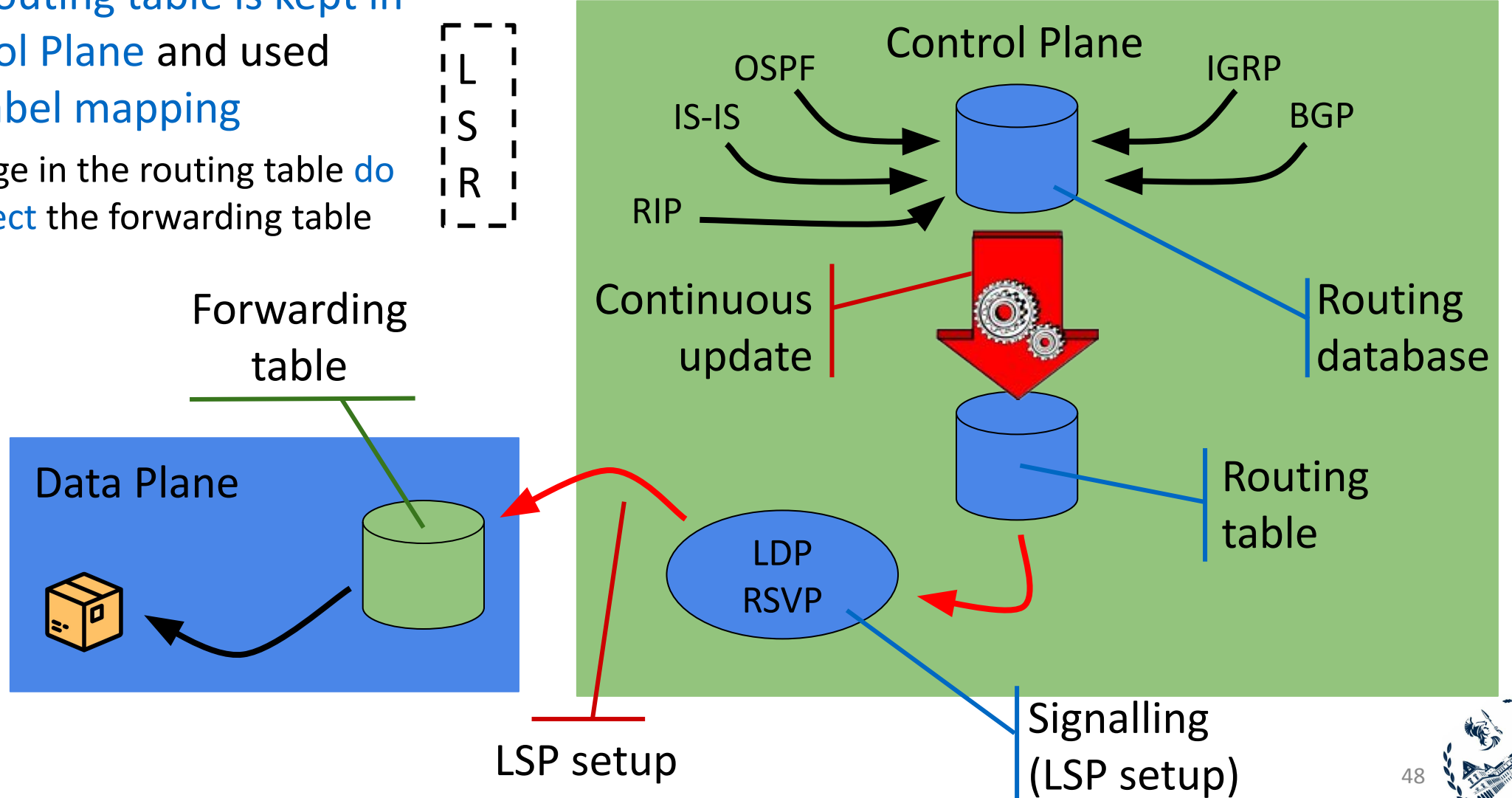
Can we do the same with IP routing?



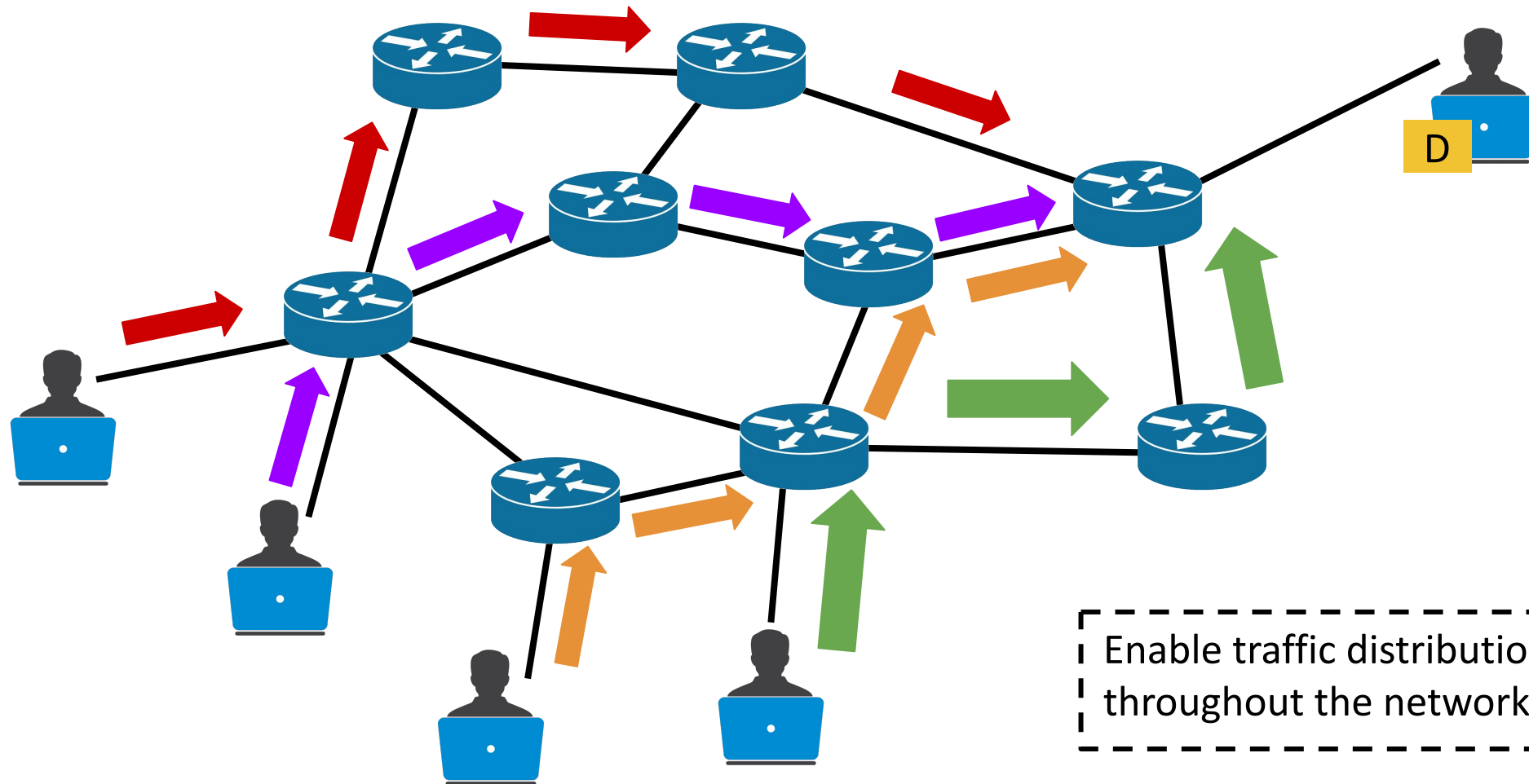
LSR Control Plane & Data Plane

- In MPLS routing table is kept in the Control Plane and used only for label mapping

- A change in the routing table do not affect the forwarding table



Traffic Engineering



Traffic Engineering before MPLS

- It was possible to have Traffic Engineering before MPLS, but only using ATM
 - But with MPLS this is done automatically, in ATM every circuit was manually (and statically) created
- Because MPLS uses the same routing protocols of IP, ATM didn't
 - It was possible to have Traffic Engineering with ATM but was not scalable
- MPLS is IP-aware



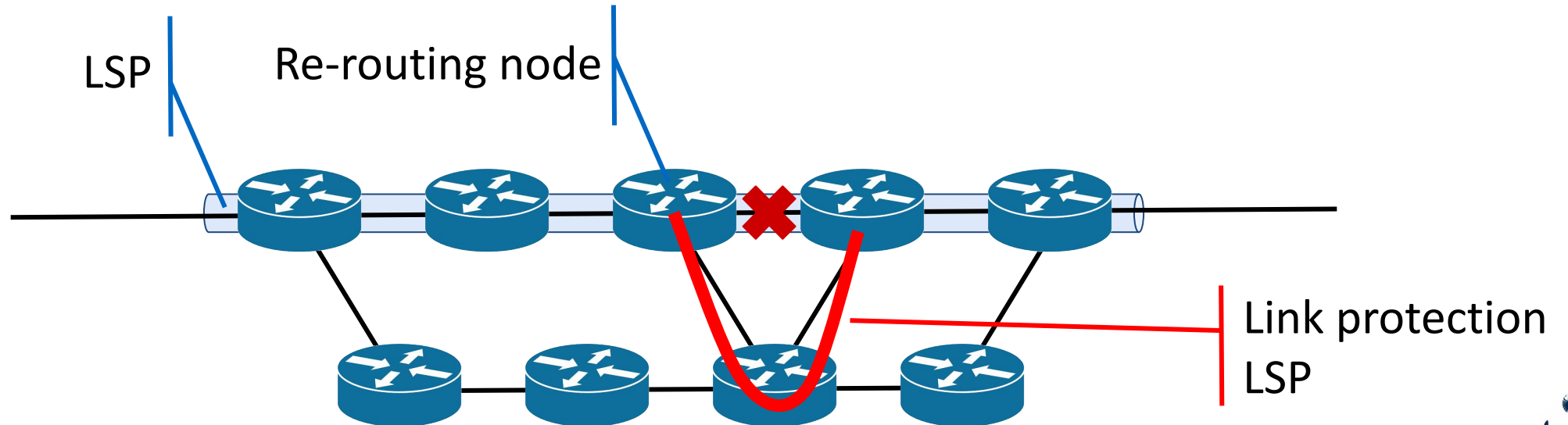
Fault recovery



Fast fault recovery

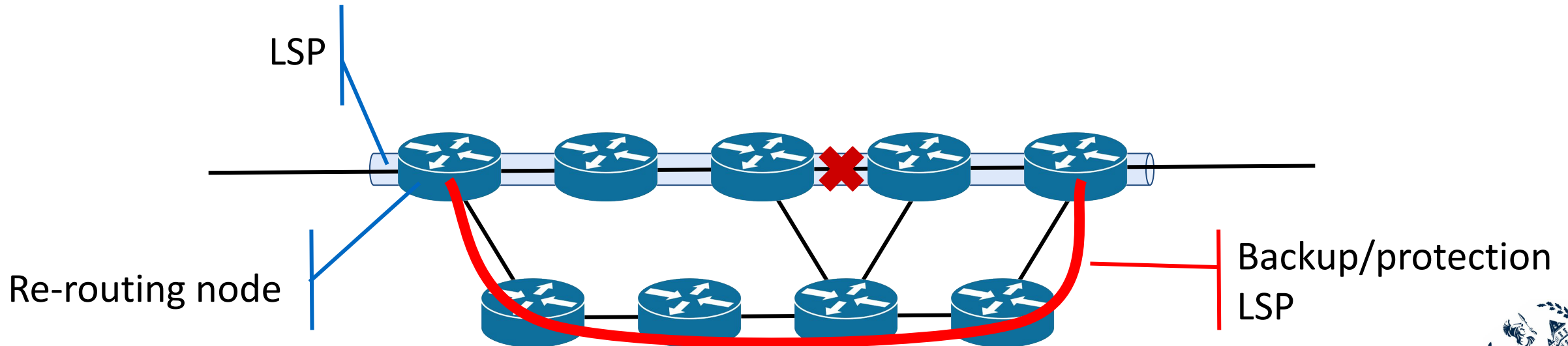
■ Link re-routing

- Creating link protection LSP for link redundancy
- In case of link fault, we will use the link protection LSP
- There will be a new label which is pushed (not swapped) for sending packets to the link protection LSP
- Very fast recovery



Fast fault recovery

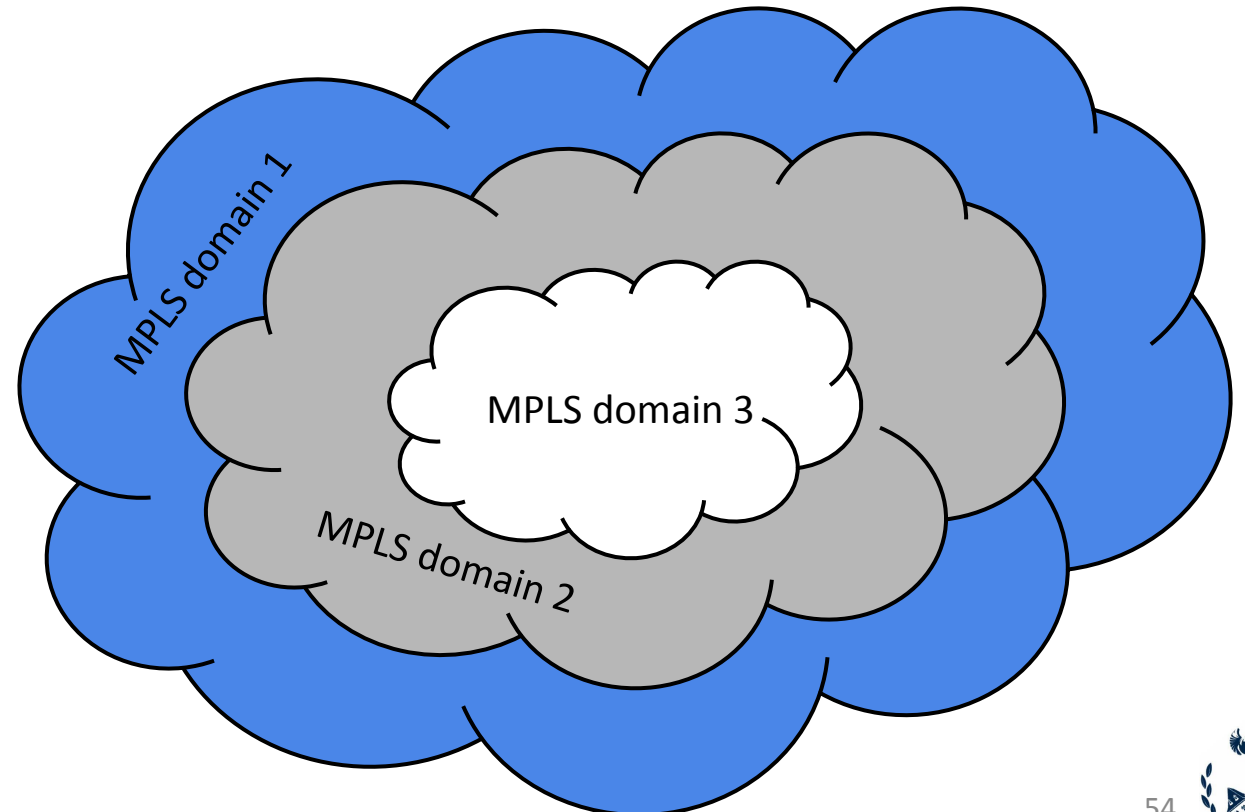
- Edge-to-edge re-routing
 - Protects the **whole LSP**
 - Works well with **explicit routing**
 - **Fault notifications** sent backwards to the re-routing node
 - The re-routing node **knows all the labels** of the backup LSP
 - **Slower** than link re-routing, **faster** than IP re-routing



Label hierarchy



- Label can be stack one another
 - Label hierarchy
 - Scalability
- Inner MPLS domain can ignore the LSPs outside and treat them as a single inner LSP with an additional label pushed
 - Reduction of routing and forwarding tables



Penultimate Hop Popping (PHP)

- There is an option in MPLS to **pop the label not at the last LSR**, but on the one to last
 - Penultimate Hop Popping (**PHP**)
- The egress LSR **do not use the label**, so we can pop it before the egress LSR
 - This **optimize the performance**
- The egress LSR **inform the upstream LSR** that he is the one to last of the LSP
- **Implicit PHP**
 - Egress LSR decides **input label 3**
 - Label 3 **means pop the label**, so one to last LSR of the LSP will pop the label
- **Explicit PHP**
 - Used when the egress LSR wants to look not at the label, but to the **other fields of the MPLS header**
 - Egress LSR decides **input label 0 which is meaningless** for MPLS protocol



Peer VPN and MPLS-based solutions



IP-based peer VPNs

■ Dedicated router

- Service provider operates a network of routers dedicated to the customer
- Viable only for major clients

■ Shared/virtual router

- Service provider creates dedicated router instances within his physical routers
- High-end routers enable hundreds of virtual routers
 - Instance-specific routing table and routing protocol
 - ASIC and operating system support
- Packet exchange through IPsec or GRE tunnels



MPLS-based Layer 2 VPNs

- **PWE3:** Pseudo Wire Emulation End-to-End
- Several services on the same network
 - IP, but also leased lines, frame relay, ATM, Ethernet
- Customer edge (CE) device features native service interface
- Traffic is carried through an LSP between CEs
- Two labels
 - External – for routing within the network
 - Identifies access point to the network
 - Internal - multiplexing of several users/services at the same access point
- There may be aggregation devices inside the network
 - e.g., an ATM switch inside the service provider network switching traffic between users
 - LSP ended on the device
- Mainly manual LSP setup
- Proposals exist for deployment of LDP and BGP



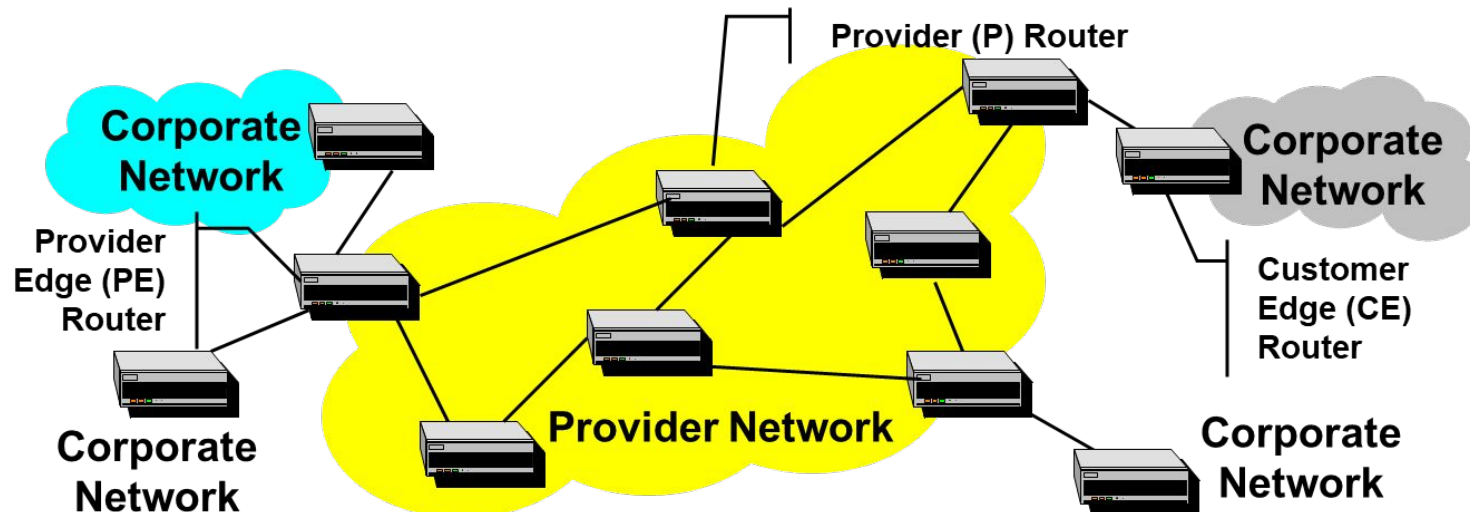
MPLS-based Layer 3 VPNs

- Provider provisioned solutions
 - VPN policies implemented by Service Provider
 - No experience needed on the Customer side
- Scalability
 - Large scale deployments
- Two alternative solutions
 - RFC 2547 bis - BGP
 - Initially supported by Cisco Systems
 - Currently most widely deployed approach
 - RFC 2917 - Virtual router
 - Initially supported by Nortel and Lucent



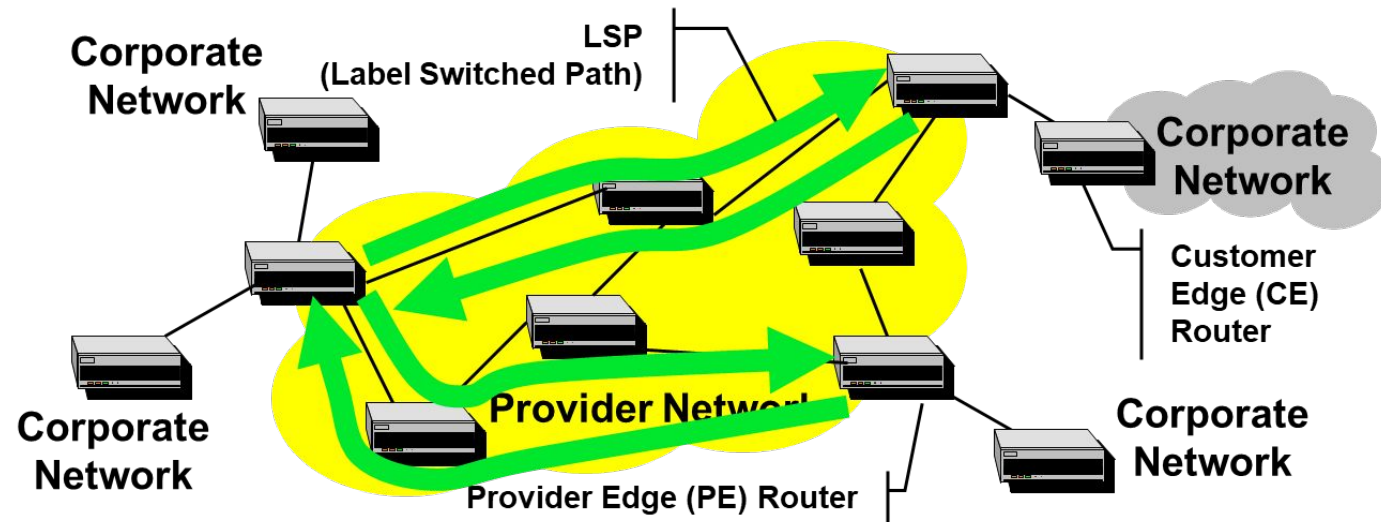
MPLS-based VPN Components

- CE router creates adjacency with PE router
 - It advertises its destinations
 - It receives advertisements of other VPN destinations
 - Static routing, or IGP (Interior Gateway Protocol)
 - (e.g., OSPF, RIP)



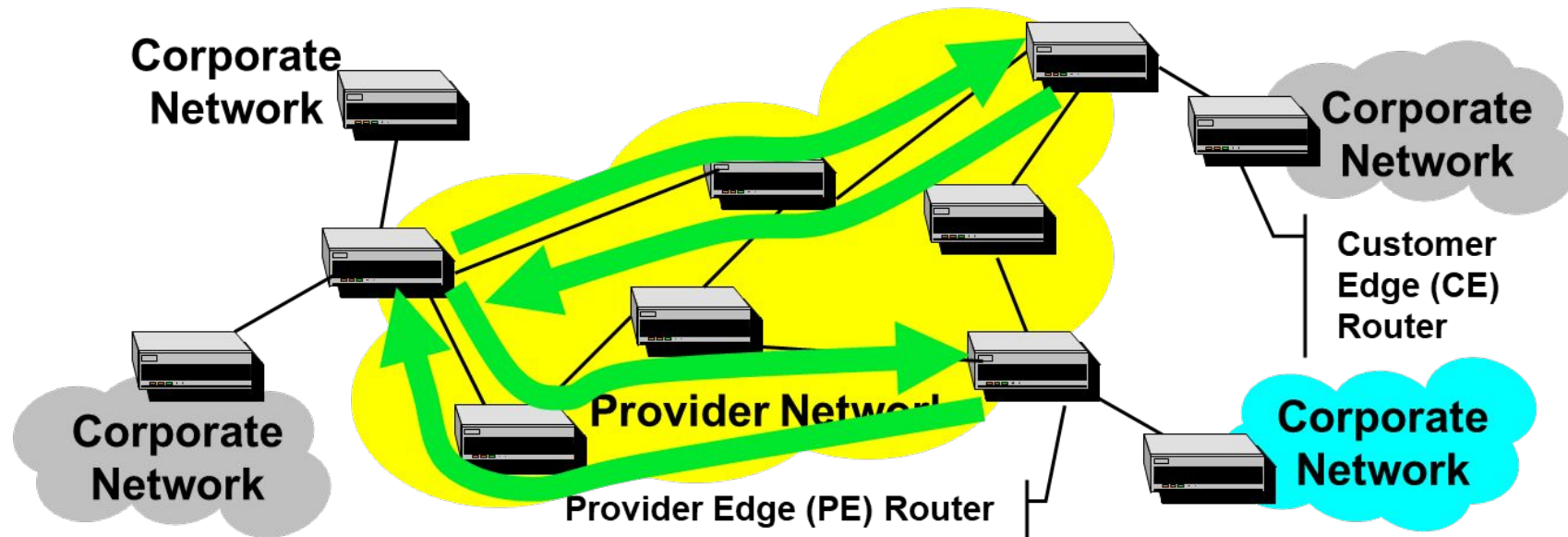
MPLS-based VPN Components

- Providers routers have routes to PE routers only
- PEs setup LSPs among themselves
 - LDP and/or RSVP (and/or I-BGP)
 - E.g., topology-based label binding



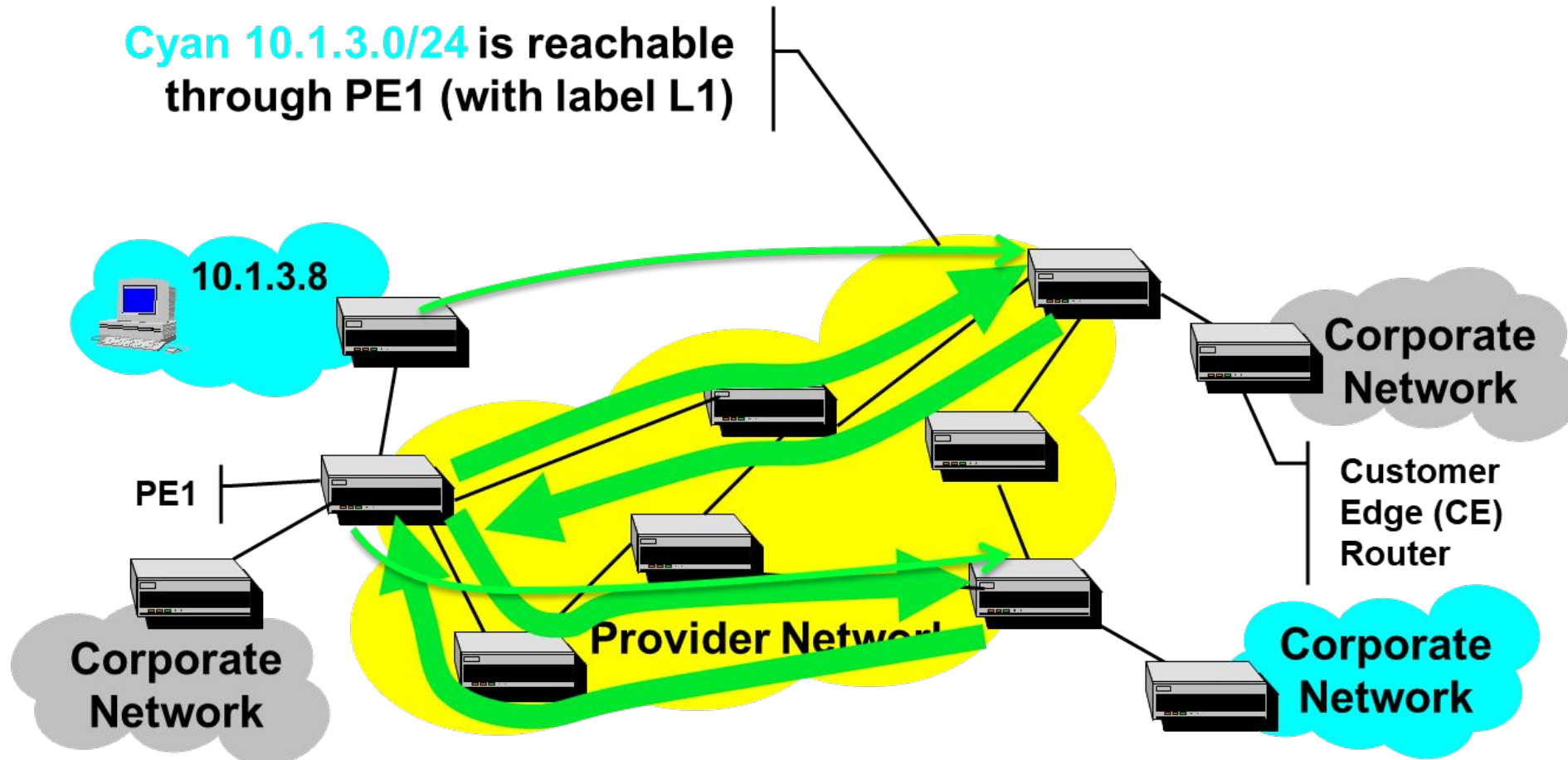
MPLS-based VPN Components

- PE routers have routes to PE routers only
 - Exchange routing information
 - I-BGP (Interior-Border Gateway Protocol) in BGP-based solution
 - IGP in VR solution
- PE keeps routes only for VPNs connected to it



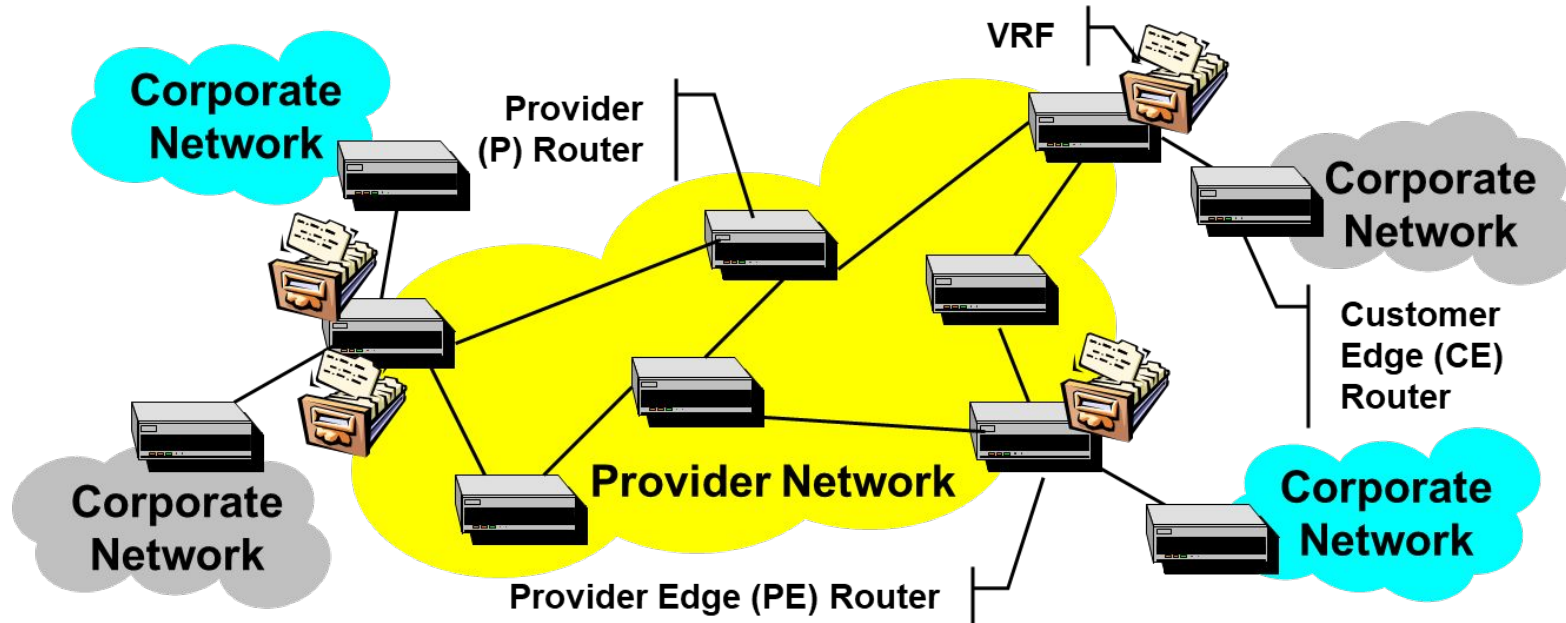
MPLS-BGP VPN Components

- PE routing exchanges (with I-BGP)



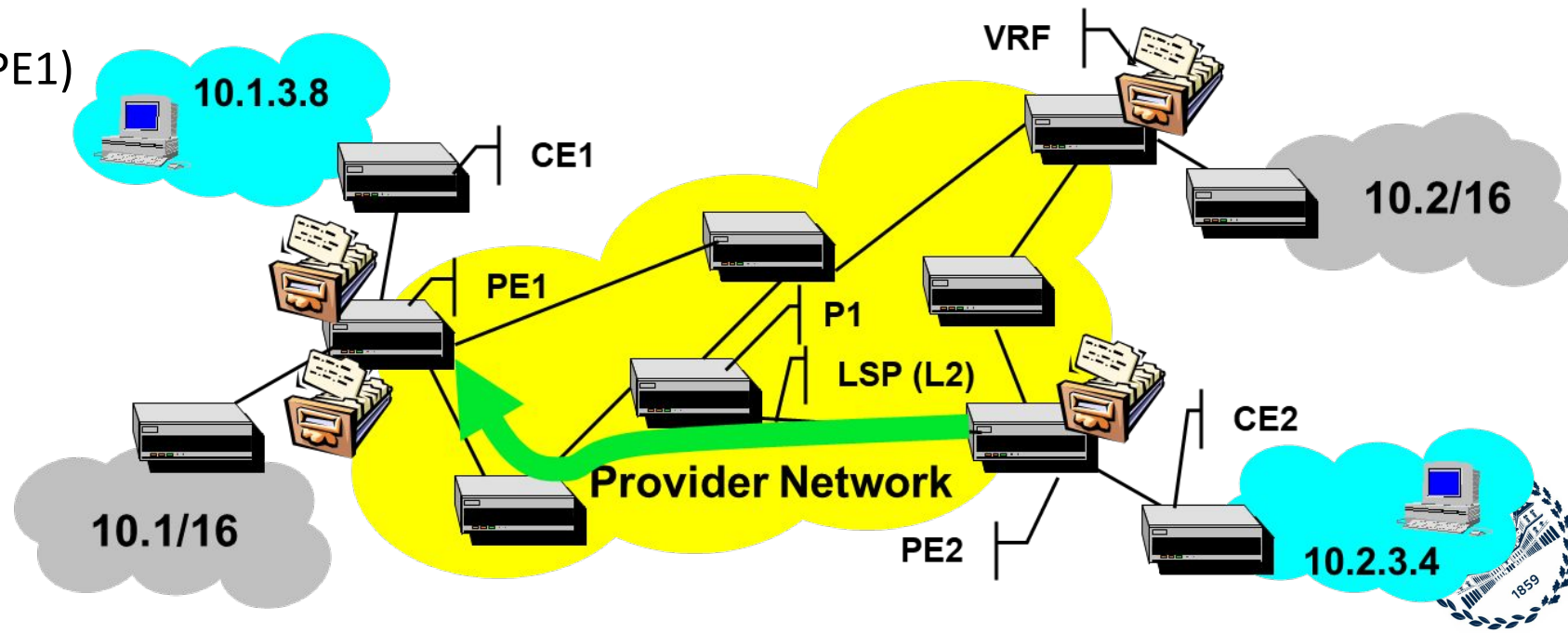
MPLS-BGP VPN Components

- VRF (VPN Routing and Forwarding) table
 - Associated to one or more (non-MPLS) ports
 - Forwarding information to be used for traffic received through the port



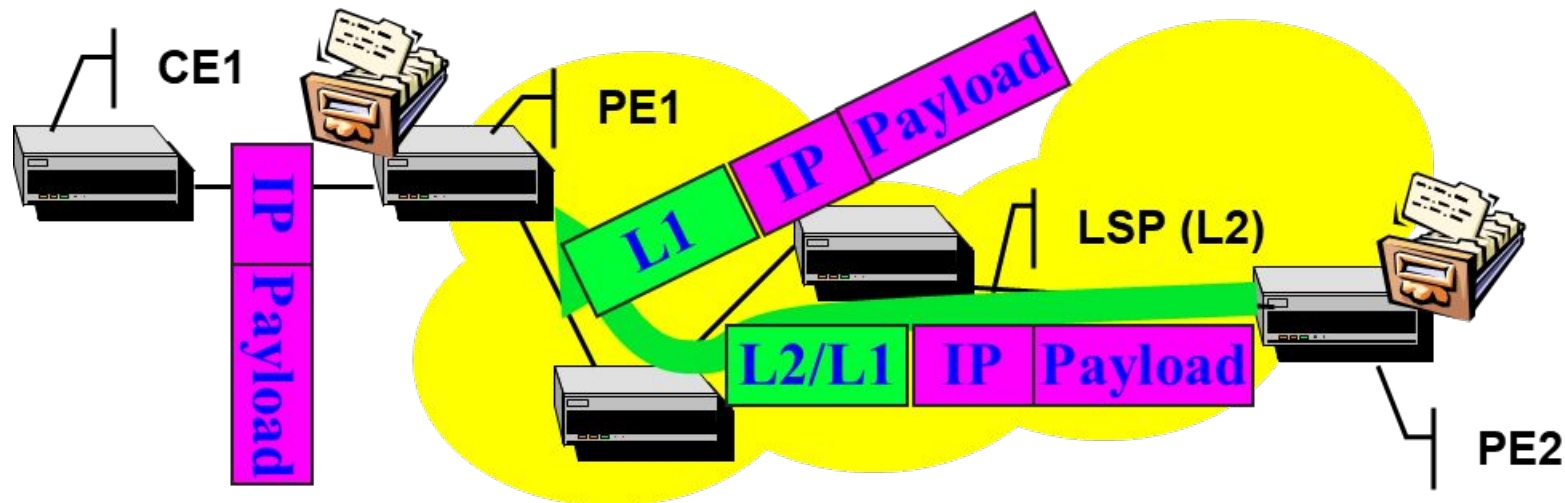
Packet routing

- Packet from cyan 10.2.3.4 to 10.1.3.8
- Default gateway -> PE2 router
- PE2 looks-up 10.1.3.8 in cyan VRF
 - Next hop: PE1
 - Label: L1 (distributed by PE1 for cyan 10.1.3.0/24)
- PE2 looks up PE1 in main table
 - Next hop: P1
 - Label: L2 (LSP from PE2 to PE1)



Packet routing

- PE2 has pushed L1 and L2 on label stack
- P routers forward packet to PE1 using L2
- Last hop before PE1 pops L2 (PHP)
- PE1 receives packet with L1
 - PE1 pops L1: plain IP packet
 - PE1 uses L1 to route packet to proper output interface



Benefits

- No constraints on addressing plan
 - Address uniqueness only within VPN
- CE routers do not exchange information with each other
- Customer does not manage backbone
- Providers do not have one virtual backbone per customer

- VPN can span multiple providers

- Security equivalent to Frame relay or ATM
 - Traffic isolation
 - No cryptography (confidentiality)

- QoS supported through experimental bits in MPLS header



Routing in MPLS-BGP VPNs

- Routing exchange at edges based on MP- BGP (Multi-protocol BGP)
 - Support for addresses of different families
- Route filtering
 - PE routers determine which routes to install in VRF
- Support for overlapping address spaces
 - VPN-IPv4 Address family
 - Route Distinguisher + IPv4 address

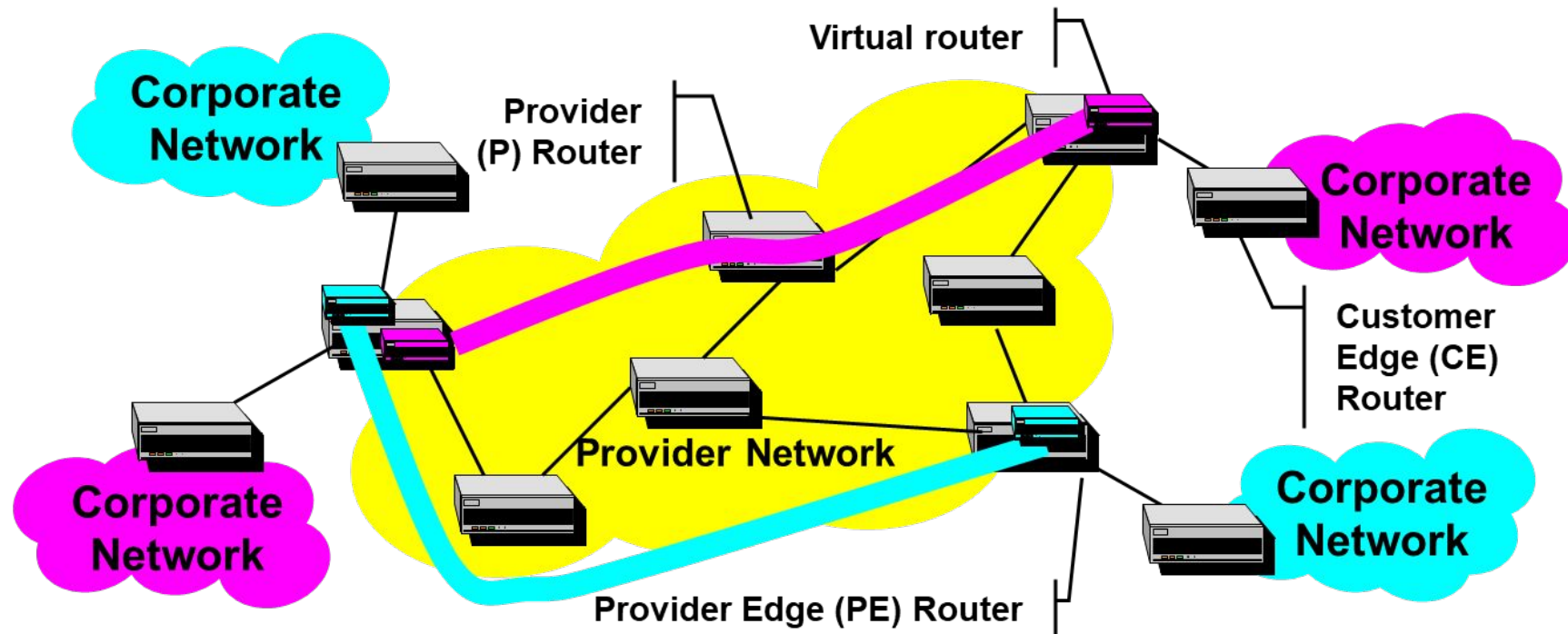
Route Distinguisher

IPv4 Address



MPLS-Virtual Router VPN

- PEs execute a (virtual) router instance for each VPN
- Each VR instance has separate data structures
- VRs of same VPN communicate through LSPs



Transport IPv6 traffic in the core of the (MPLS) network



DS-Lite and IPv6 backbones

- **DS-Lite** assumes that the backbone can transport IPv6 traffic natively
- For instance, IPv4 transport is not even required
 - IPv4 traffic is in fact tunnelled toward the Address Family Transition Router (AFTR)
 - IPv4 routing is completely useless in this particular scenario
- Question: would any sane provider turn its backbone into IPv6-only network?
 - Are we going to abandon the working IPv4 backbone for a brand new (unknown, unexperienced, not well tested, etc) IPv6 backbone?



6PE

- 6PE may offer a nice answer to any NSP whose core network runs MPLS
- Idea
 - Keep the core unchanged
 - (not excluding that it can evolve in the future)
 - Add support for IPv6 networks at the edge
 - networks giving IPv6 connectivity to end users
 - Distribute IPv6 routing information in MPLS/BGP, in the same way as we currently do with VPNs
- [RFC 4798] J. De Clercq , D. Ooms, S. Prevost, F. Le Faucheur, “Connecting IPv6 Islands over IPv4 MPLS Using IPv6 Provider Edge Routers (6PE),” February 2007.

Possible options for IPv6 over MPLS networks

- (1) Native IPv6 over MPLS
 - IPv6 and IPv4 traffic is treated identically by the core routers
 - Core Infrastructure requires full Control Plane upgrade to IPv6
 - IPv6 Routing in core
 - IPv6 Label Distribution Protocol in core
 - Dual Control Plane management if IPv4 and IPv6 services
- (2) IPv6 over Circuit Transport over MPLS
 - L2 frames, e.g. Ethernet frames or ATM cells, are encapsulated into MPLS frames and transported over the network
 - No changes are needed to Provider routers
 - Scalability problems arise in heavily (L2 tunnel) mesh topologies
- (3) 6PE
 - Similar to MPLS VPNs in terms of technical implementation and complexity



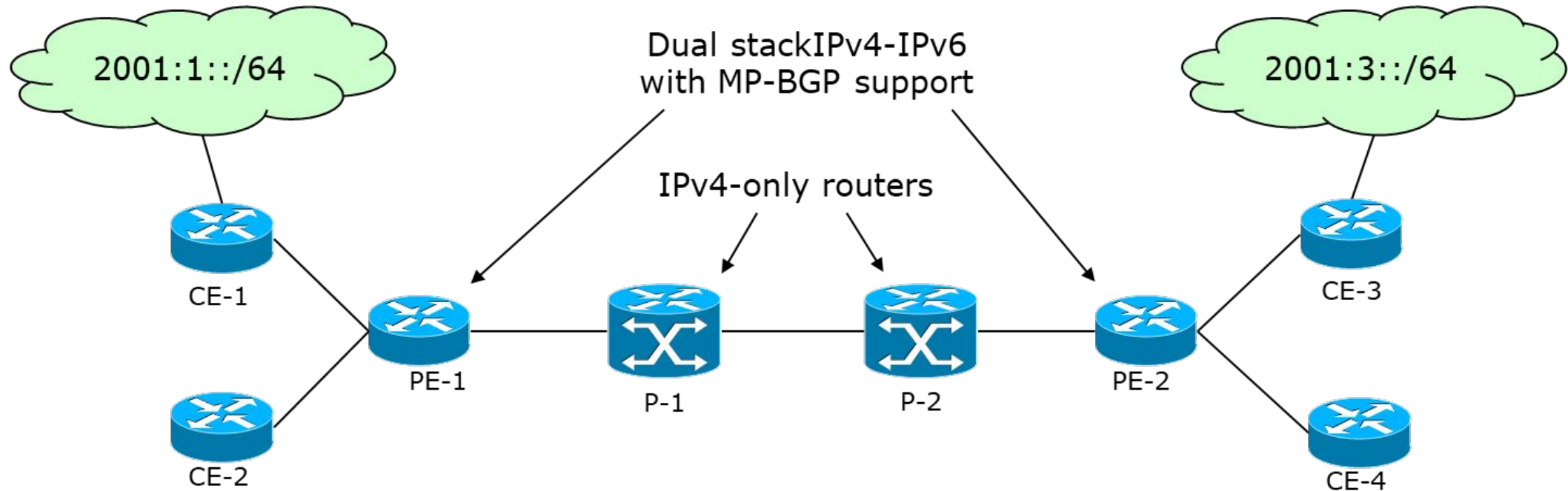
6PE: applicability

- 6PE is a good choice for ISPs that have MPLS core network and supports MPLS VPN (or other) services
- IPv6 services are requested by a limited number of customers
 - If the number of IPv6 customers increases so that the ISP requires most of the access routers to become 6PE, the NSP should consider to upgrade to whole network
- The ISP wants to avoid either to fully upgrade the core network or to deploy IPv6-over-IPv4 tunnels



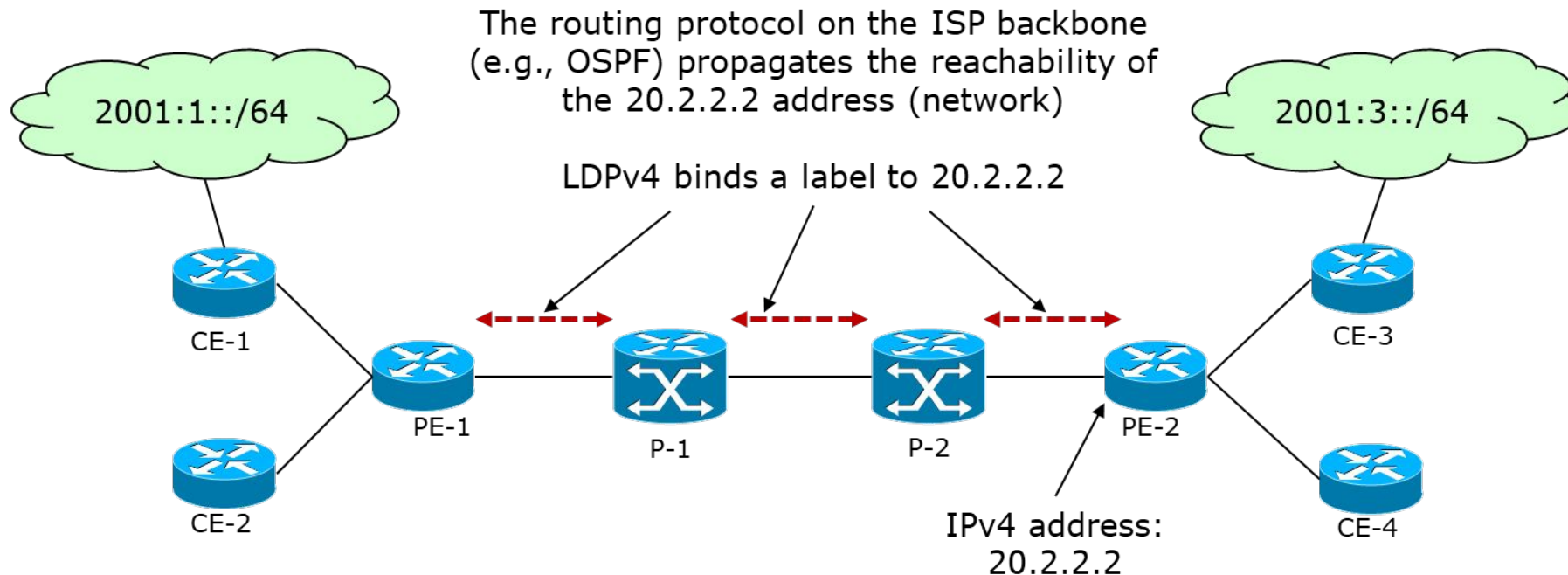
Requirements

- The ISP has to upgrade the Provider Edge (PE) routers to support IPv6 and MP-BGP
- Core routers (P) do not need any change in terms of configuration or software



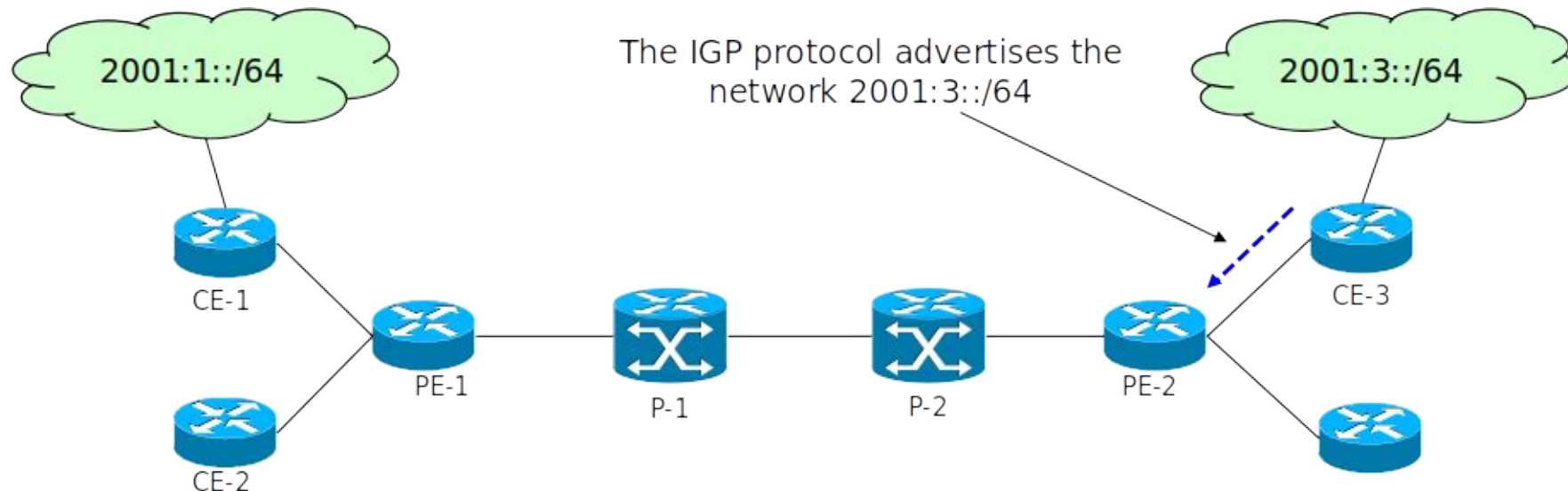
Announcing IPv6 networks

- 6PE routers advertise their IPv4 address (network) into the IGP routing protocol of the IPv4/MPLS core network
- Each router in the MPLS domain will eventually assign a label corresponding to the route for each 6PE router



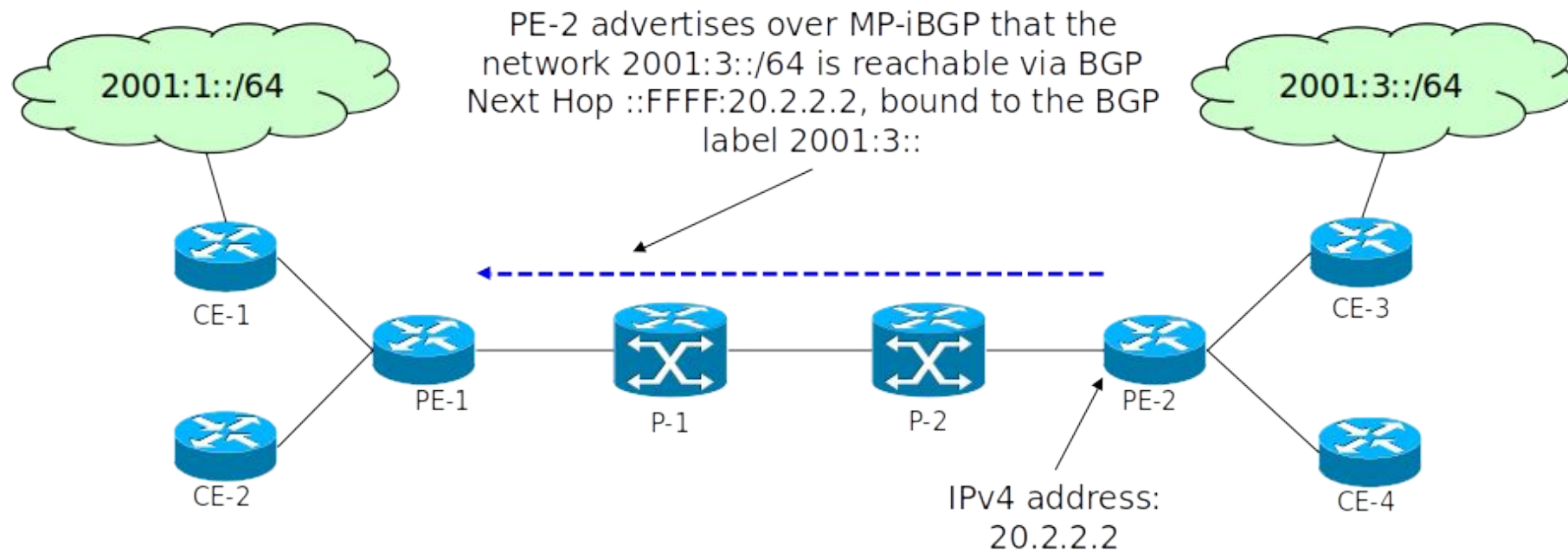
Announcing IPv6 networks

- Customer Edge (CE) routers and 6PE routers are connected with (one or more) logical or physical native IPv6 interfaces
- Any common routing protocol (e.g., OSPF, eBGP, static or default routes) between CE and PE allows the distribution of IPv6 reachability information



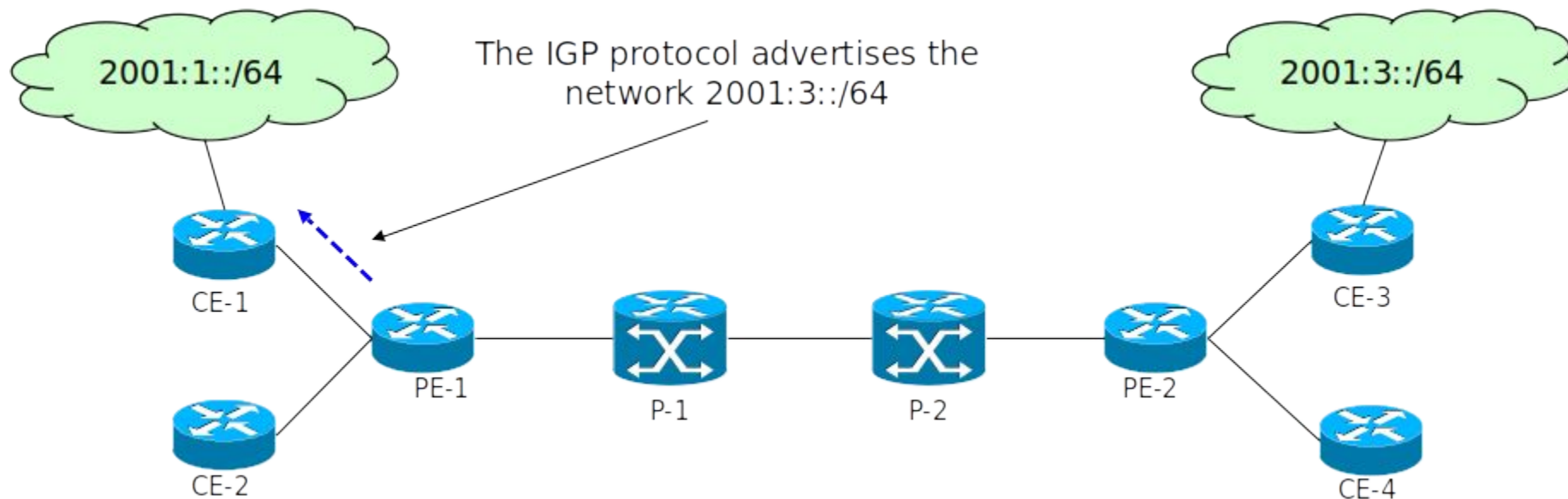
Announcing IPv6 networks

- Customer IPv6 prefixes are exchanged among the 6PE routers over MP-BGP session running over IPv4
- 6PE routers convey their IPv4 address as the BGP Next-Hop for the IPv6 prefixes
 - The BGP Next Hop is the IPv4-mapped IPv6 address of the 6PE



Announcing IPv6 networks

- Finally, the IGP protocol on the other side of the network propagates the advertisement within the network of the other customer
 - This may not be needed e.g., in case a default route is used



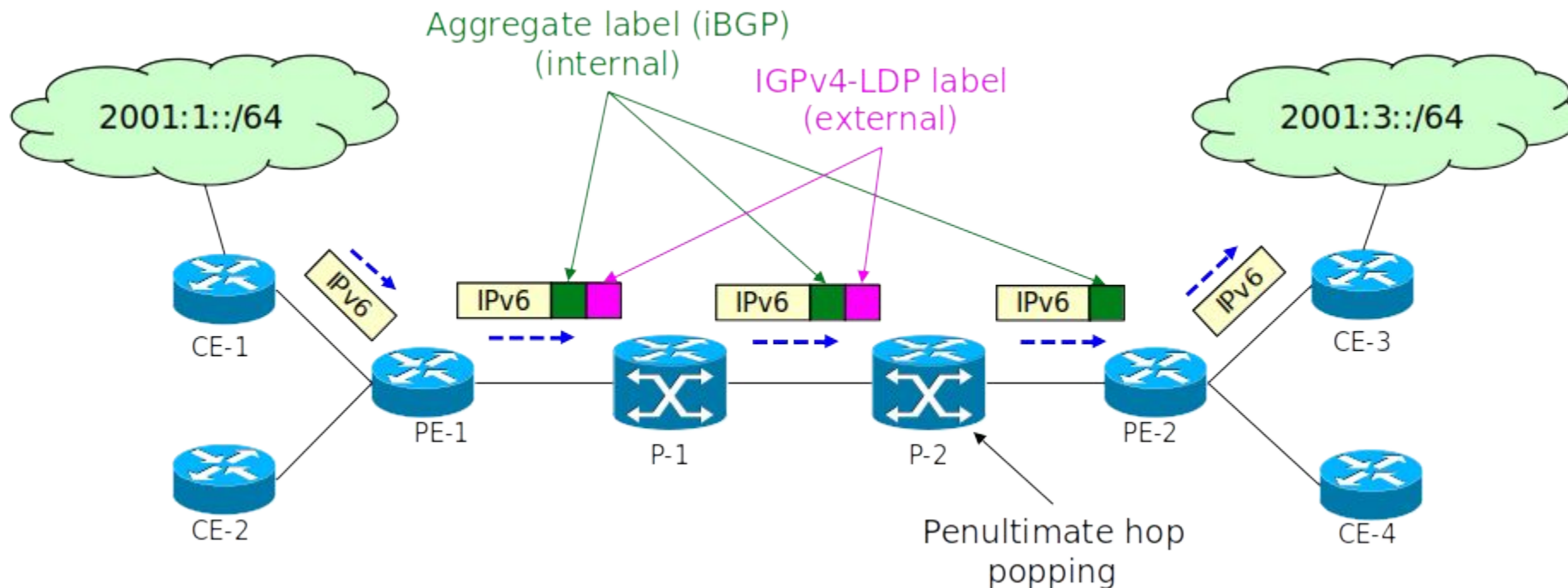
Distributing MPLS labels

- Label distribution and setup is done differently for IPv4 and for IPv6 routes
 - IPv4 labels are distributed through LDPv4 in the core network (P routers)
 - IPv6 labels are distributed with MP-iBGPv4 among the PEs



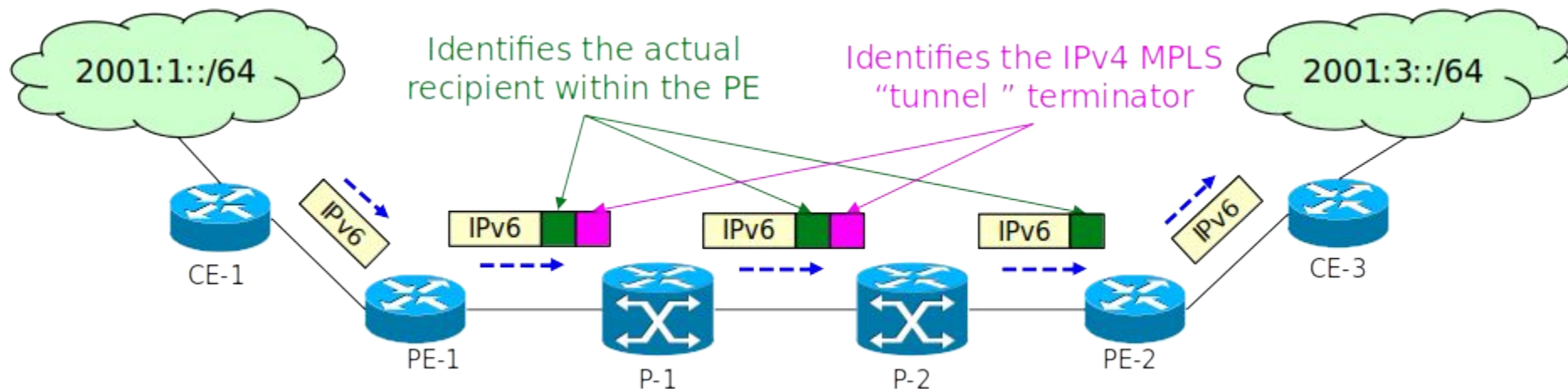
Forwarding IPv6 traffic

- The CE sends a IPv6 packet to PE-1
- The ingress 6PE router tunnels the IPv6 data over an LSP towards the Egress 6PE router identified by the IPv4 address that derives from the IPv4-mapped IPv6 address of the BGP Next Hop field for the corresponding IPv6 prefix

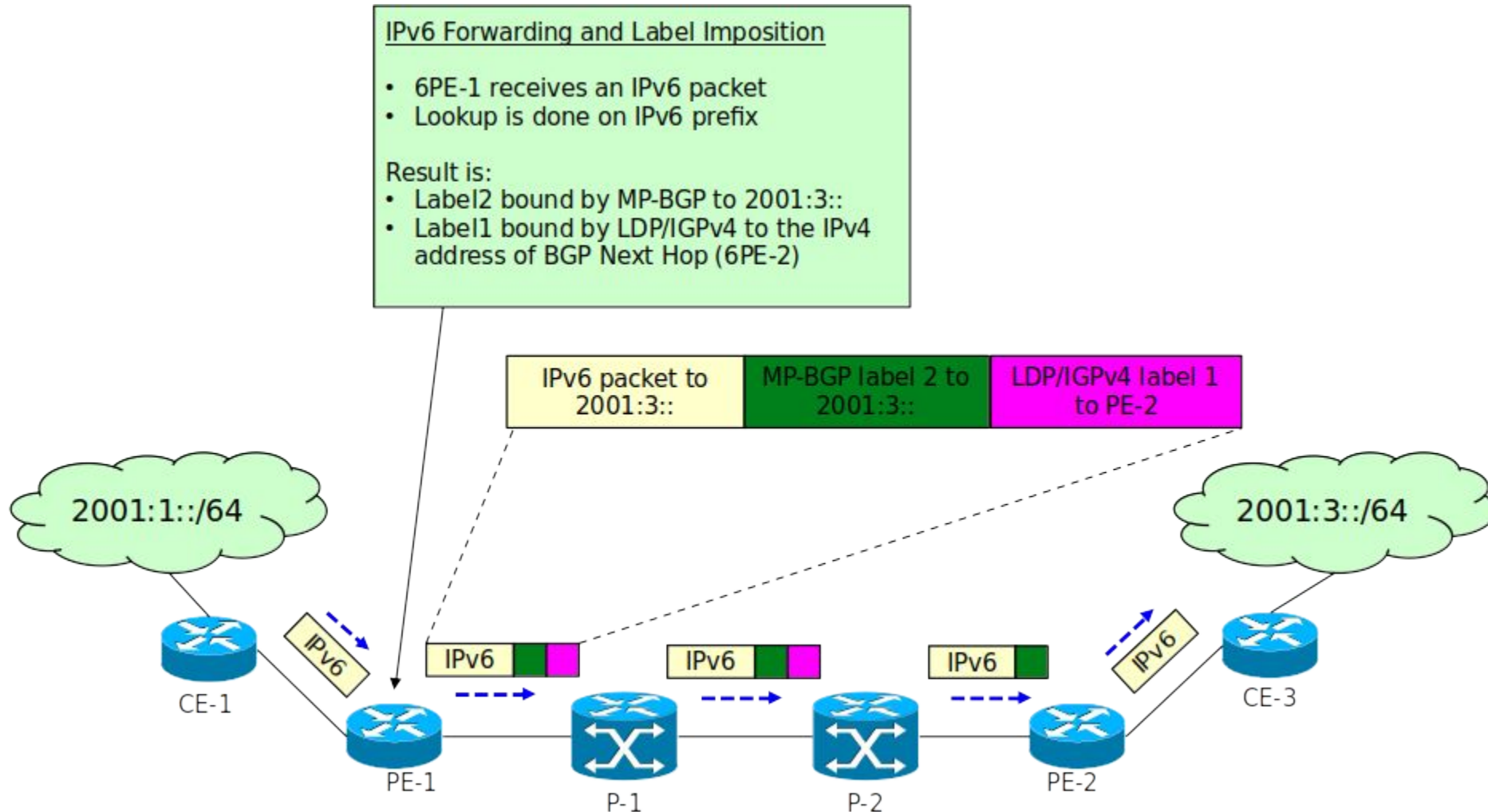


Forwarding IPv6 traffic

- 6PE encapsulation involves two labels
 - The inner label (“aggregated IPv6 label”) is bounded to each advertised destination IPv6 prefix
 - The outer label is associated with the egress 6PE IPv4 address
- Note that the inner label is in principle not required for the operation, but it keeps the solution the same as for other tunneling, e.g., VPNs
 - Without the inner label the “penultimate hop” router would have to be able to forward a plain IPv6 packet to the egress 6PE router



Forwarding IPv6 traffic



Conclusions

- PE routers have to be dual stack and to support MP-BGP, while P routers do not need any modification
- Provide native IPv6 services to customers without changing the IPv4 MPLS core network, which turns into minimal operational cost and risk
- 6PE scenario is similar to packet forwarding in MPLS VPNs
 - IPv6 CEs have a single routing peer and do not need any change whenever remote IPv6 CEs are connected or removed
- 6PE fits very well into the general MPLS philosophy
 - However, 6PE does not justify the deployment of MPLS core network
 - Therefore, 6PE is meant for deployment scenarios where the MPLS core is available

